

Gradland: On Phenomenal Experience, Differentiated Across Many Dimensions

David Balduzzi

Abstract. This paper investigates the hypothesis that the first-order structure of physical interactions, i.e. gradients or Jacobians, characterizes the structure of phenomenal experience. It does so in an idealized world inhabited by neural networks, Gradland, where the physics are known and the functions are (mostly) differentiable. The paper introduces two measures of Jacobian structure: effective rank and cohesion, based on Kirchhoff complexity. Applying the measures to a series of worked examples shows the hypothesis accounts for: (1) the duration of experience, that it can prolong over hundreds of milliseconds; (2) the difference between what is experienced vividly and obscurely; (3) the experience of texture; (4) the blooming buzzing confusion presumably experienced by newborns; (5) the difference between ideas that are held distinctly in mind and ideas that are confused; (6) what learning is like; and finally (7) the paper explains the function of rich, dense experience.

Code: github.com/dbalduzzi/gradland

1 Introduction

How do things hang together [1]? To tackle the question for physical phenomena, Leibniz and Newton invented calculus and initiated the most successful research program in history: studying nature via differential equations. Centuries later the branch of that research program called applied mathematics has three main workhorses: linearization (differentiation), aggregation (weighted averages, matmuls, discrete integration) and diagonalization (eigenvalues, singular value decomposition). I will apply these tools to study how phenomenal experience hangs together.

The premise is that phenomenal experience is generated by physical interactions: interactions between photons, atoms, neurons, eyes, ears and trees. But how to relate phenomenal experience to physical structure? We will spend time analyzing gradients in abstract models of physical systems and see what that buys us. A lot, it turns out.

Why these tools and why now? Because modern AI is surprisingly powerful and is also, surprisingly, built on derivatives and matmuls. Now is a good

time to ask what happens if we take derivatives and matmuls as fundamental building blocks – rather than things, such as atoms and cells, or tools that are less central to mathematics, such as logic and information theory. Although gradients have been ubiquitous in science and engineering for centuries, their long-range propagation and high-dimensional interaction were not intensively studied until recently. The field of deep learning contains within itself a science of gradients and a proto-science of interactions.

If derivatives and matmuls are so powerful, why are they not used to model experience already? I can only guess. Matrices are arguably the simplest model of interactions. They are daunting in practice only because they are so often so big. The world is not linear. Right there at the start you have thrown away a lot of detail and it is not coming back. The same goes for derivatives. Modeling experience with derivatives is like attending a seminar by an economist who understands the marginal utility of everything and the fair value of nothing. Why put yourself through that? We are going to do it, and it will be worth it.

1.1 Overview

Mach proposed that matter and mind, the physical and the psychical, are two perspectives on reality [2]. I prefer the terms physical interaction and experience, but will stick with matter and mind for this subsection. The paper instantiates Mach’s proposal as Hypothesis G: the hypothesis that the first-order structure of matter characterizes the structure of mind §3.2. This subsection introduces the main ideas.

Matter and mind are extrinsic and intrinsic perspectives on reality: you can look *at* a brain or look *as* a brain. To look *as* a brain is to see a tree, say, through your visual cortex, through your eyes, and through the interjacent atmosphere. To see *as* a physical system is to experience what matters to it – or what makes a difference to it. To what is it sensitive? Derivatives are perhaps the simplest way to study a system’s sensitivities to the world (its inputs) and to its own internal structure (its parameters).

The first observation is that mind is, or can be, *transparent*: to look *as* a brain is to see *through*

matter. Not just through air, but more interestingly through your eyes and neurons as well. Transparency is not a binary yes/no phenomenon and is not just about what you see, although I lean heavily on visual and optical metaphors §2.2. You see, feel, smell, and taste *many* things by means of your brain, nervous system and body. Transparency gets at both mind’s compositional structure and, in humans at least, its relatively high capacity. The language of gradients and matrices is especially convenient when studying interactions, §1.2 and §2.3. Matrices combine, or rather compose, to form yet more matrices by matrix multiplication, which is also, by the chain rule, how gradients compose. The capacity of matrices can be quantified using their singular value spectrum §3.3.

The second observation is that mind is *cohesive*. Mind *extends* over time and across space: it does not localize in a single gland, neuron, atom or sub-atomic particle. Some kind of glue binds neuronal activity together, from its own perspective, over many milliseconds and across many millimeters. The glue can only be the structure of the interactions themselves. Cohesion quantifies the density of the interactions that span all the inputs *to* and outputs *from* matrices §3.3. If components of a system do not interact, then the system is not a system but rather many non-interacting systems. Cohesion can be applied to analyze shared experiences, see §4.1, and our experience of our immediate past, see §4.2.

The paper studies the experience of neural nets in Gradland, which abstracts over the details of hardware implementation. It is organized as follows. §2 provides motivations for some aspects of the approach. §3 introduces notation, unpacks Hypothesis G, and presents the notions of transparency and cohesion. The implications are worked out via a series of examples in §4, which discusses interesting aspects of neural nets, and §5, which is concerned with learning, the function of experience, and how experience hangs together in neural nets.

1.2 Related work: Why gradients?

The approach taken here is derived from integrated information theory v2 [3], which I codeveloped with Tononi. The various integrated information theories [3–5] are closely related, so it is worth explaining some key differences; see §A.6 for detailed comparison.

The initial source of divergence, with many knock-on effects, is using gradients and matrices instead of information theory. Integrated information (all versions, not just v2) is limited to toy examples because the tools do not scale. Gradients are cheaper and more flexible than information-theoretic measures. They are battle-tested tools that scale well.

Their usefulness in physics and elsewhere gives some confidence that they capture something real.

There is a weak analogy between gradients and information. An AND-gate maps inputs $\{00, 01, 10\} \rightarrow \{0\}$ and $\{11\} \rightarrow \{1\}$. If an AND-gate outputs 0, that tells you less than if it outputs 1 because 0 rules out fewer alternatives. Another way to say the same thing is that the path leading to $\{1\}$ is more sensitive to input perturbations than the path leading to $\{0\}$.

Gradients also quantify the sensitivity of outputs to changes in inputs. They are relatively cheap because instead of considering all possible inputs, they only consider infinitesimal perturbations of the actual input. The computational cost of cohesive rank is $\mathcal{O}(nd \cdot \min(n, d))$, see §3.3, with lots of opportunities for improvement, whereas integrated information is doubly exponential, see §A.6.

A difference worth mentioning is that, whereas (some version of) integrated information is zero for feedforward networks [6], cohesive rank is not, §4.2. Arguably, *everything* is feedforward in spacetime; at least, it seems ill-advised to systematically rely on closed timelike curves.

1.3 Boundary problems

An important lesson implicit in modern AI is that intelligence is hard to measure. Frontier labs have long generated spreadsheets worth of benchmarks [7]. They can pull in different directions [8]. A model is rushed into production because of coding improvements – but then chat users experience a downgrade. Models exhibit jagged capabilities *despite* the best efforts of labs to smooth them out [9]. Imagine if they optimized for lopsidedness. The space of capabilities is high dimensional and strange.

“These four options are logically exclusive and exhaustive. Either nothing is conscious (eliminativism), or everything is conscious (panpsychism), or there’s a sharp line between conscious and nonconscious systems (saltation), or there are borderline cases (indeterminacy).” —Schwitzgebel [10]

Papers on consciousness often elide definitions and skip straight to talking about human consciousness: “what abandons us every night when we fall into dreamless sleep and returns the next morning when we wake up” [11]. Starting with humans and looking at the descent of man, it is then easy to believe that conscious minds can be quantified and ranked: higher for us, lower for apes, and down it goes. Every one’s consciousness and even existence ordered from top to bottom [12]. But global rankings at best partially capture capabilities and make no sense for consciousness.

I defer value judgements as long as I can. Instead of human consciousness, I start from the hypothesis that experience is first-order interactions. I study its implications in Gradland, a world that does not exist [13], whose inhabitants, abstract neural nets, are not conscious – *if by conscious you mean something that might suffer, something we should think twice before switching off*. The space of neural nets is vast and most of it is utterly alien. There are endless interesting phenomena. However, ethical considerations cannot be postponed forever. They arrive, long after experience, shortly after metabolism, in [14].

Experience is real. Its varieties can be measured in many ways. As for boundaries, they are indeterminate in general. The space of experiences is high-dimensional and parts of it are beyond bizarre. Let's study the structure of experience. The experience of time passing. How learning changes experience. Not ranking nor boundaries.

2 Motivations

The motivations below provide three complementary paths into Gradland.

2.1 Goff's experience inverts

Experience is generated by physical interactions. Equivalently, experience is generated by behavior where behavior, potentially but not necessarily, goes all the way down and across [15–18]: behavior is what your neurons are doing, but also your glia, your eyes, your hands, and the tree you are looking at. Behavior is interactions and experience is what it is like to interact.

The alternative is to allow experience inverts:

"If it makes sense to separate out consciousness and behavioral functioning, then it also makes sense to mix and match them in weird ways. We could imagine, for example, [...] pain-pleasure inverts who behave just like us but feel pleasure when we feel pain, and vice-versa. When you stick a knife in a pain-pleasure invert, they feel great pleasure, but this causes them to scream and run away." —Goff [19].

Experience inverts are a useful *reductio ad absurdum*: a complete rejection of any relation between experience and behavior. If we accept physical interactions without experience, that is philosophical zombies [20], then we should also accept experience inverts – at which point we are open to wondering if the wind feels angst or the rain is bored. This is a dead end.

2.2 Bergson's convergent lens

It is famously hard to imagine phenomenal experience arising from mechanical systems:

"One is obliged to admit that perception and what depends upon it is inexplicable on mechanical principles, that is, by figures and motions. In imagining that there is a machine whose construction would enable it to think, to sense, and to have perception, one could conceive it enlarged while retaining the same proportions, so that one could enter into it, just like into a windmill. Supposing this, one should, when visiting within it, find only parts pushing one another, and never anything by which to explain a perception."
—Leibniz [21]

To counter the image of Leibniz' mechanical mill, consider the effect of a convergent lens. One way to study color is to start with the set of colors: $\text{color} = \{\text{blue}, \text{red}, \text{yellow}, \text{cyan}, \dots\}$. This is familiar but lifeless. By construction the set ignores precisely what differentiates one color from one another. The alternative is to start from how colors interact:

"Let us imagine, for example, all the colors of the rainbow, violet and blue, green, yellow and red. [...] There are two ways of determining what they have in common. The first consists simply in saying they are colors. The abstract and general idea of color thus becomes the unity to which the variety of shades is reduced. [...] Quite different is the method of true unification. In this case it consists in taking the thousand and one different shades of blue, violet, green, yellow and red, and, by having them pass through a convergent lens, bringing them to a single point. Then appears in all its radiance the pure white light which, perceived here below in the shades which disperse it, enclosed above, in its undivided unity, the indefinite variety of multi-colored rays." —Bergson [22]

Bergson is not quite right because there is more to color than the frequency of light. Let's take his lens seriously but not literally. Instead of comparing minds to mills, clocks, telegraphs, switchboards, and computers, imagine lines of force pulling, pushing and intertwining [23]; interacting matter and radiation, converging and diverging rays, deformations, interference, polarization, transformations and caustics.

2.3 Heisenberg's matrix mechanics

Why matrices? The short answer is that derivatives are linear transforms, and matrices are what linear transforms look like when you fix a basis. A more interesting answer goes back to the founding document of quantum mechanics:

“From this state of affairs, it seems more advisable to give up completely on any hope of an observation of the hitherto unobservable quantities (such as the position and orbital period of the electron), and [...] to attempt to construct a quantum-theoretical mechanics [...] in which only relations between observable quantities would be present.” —Heisenberg [24]

Heisenberg considered only what is observable; organized observables into tables of numbers (‘matrices’, although he did not know the word); and modeled how observables evolve over time by manipulating these tables of numbers (using matrix multiplication, which he reinvented). In short, matrices, representing observables and their relations, are the fundamental components of Heisenberg’s quantum-theoretical mechanics rather than particles or trajectories. Physics has progressed since 1925 but let’s take this core idea, that matrices are fundamental.

The operators in quantum mechanics have additional structure that, for example, allows to compute probability amplitudes. This structure does not scale; the matrices used to model macroscopic phenomena have no such structure and there is no hope of reducing macroscopic models to quantum calculations. But, it is a hard-won, empirical fact that linear approximations (matrices and their ilk) are useful if not primary tools when studying reality at most scales. To be clear, the claim is not that reality is matrices all the way up and down. Rather, (1) we should focus on relations rather than things and (2) start with matrices because they are ubiquitous tools for representing relations.

“The relations between things, conjunctive as well as disjunctive, are just as much matters of direct particular experience, neither more so nor less so, than the things themselves.” — James [25].

There is even less hope of directly observing the phenomenal experience of a third party than there is of observing the orbital period of an electron. Following Heisenberg, let’s study phenomenal experience based on relations between observable quantities. However, unlike in physics, let’s not attempt to find the base layer of reality. We work *in media res*.

3 Gradland

This section introduces notation, presents the hypothesis, and provides tools for analyzing Jacobians. Gradland is a rebranding of the mathematical models *actively used* by AI researchers. In Gradland systems are (mostly) differentiable and perturbations are infinitesimal. Inputs and outputs are observed. The

underlying physics, that is the functions mapping inputs to outputs, is known. There is one small novelty. In Gradland, Hypothesis G, that experience is first-order interactions, is assumed to be true.

To prevent readers with a machine learning background from starting on the wrong foot, note that *for the primary purposes of this paper, loss functions and their gradients are not privileged*, see in particular §4.4. They are important examples and play a more prominent role in §5.

3.1 Notation and basics

The notation that follows is overkill; it gives the flavor of the structure underlying neural nets. There is a directed acyclic graph (DAG) G . Nodes of G are marked by a spacetime coordinates (j, t) . Spatial coordinates j refer to neurons. Temporal coordinates t are so we can talk about neural activity at different times. Each neuron has a parameter vector $\mathbf{w}_{(j,t)}$, which can change with time but won’t until we consider learning in §5, and an activation function $f_j : \mathbb{R} \rightarrow \mathbb{R}$. The output at (j, t) is $f_j \left(\mathbf{w}_{(j,t)}^\top \cdot \mathbf{x}_{\bullet \rightarrow (j,t)} \right)$ where $\mathbf{x}_{\bullet \rightarrow (j,t)}$ collects neuron j ’s inputs at time t into a vector.

Given a graph G , define a **system** as a subgraph S , with some nodes of S labeled as inputs and outputs. The inputs and outputs of S should be nonempty and disjoint. Examples of outputs are: the loss function; the outputs of a particular layer; or all outputs of all layers except the first (assuming it is the input). The parameters of input nodes are ignored – as if they do not exist. Let $\mathcal{W}$ denote all parameters in a system.

The specification of inputs and outputs matters. A pathological example is to put outputs *upstream* of inputs. The Jacobian is the zero matrix; the system is inert. The measures in §3.3 will return zero.

Examples.

1. *Multi-layer perceptron (MLP)*. There are $L + 1$ layers of $\{d_l\}_{l=0}^L$ neurons. There is a connection from every neuron in layer l to every neuron in layer $l + 1$. The spatial coordinate now has two dimensions: the layer and position in the layer. Assuming T time points, the graph $G(\text{MLP})$ is T disconnected subgraphs. Each subgraph is a copy of the diagram of the MLP. Subgraphs are disconnected because there is no interaction across time (again, there is no learning for now).

Suppose all neurons in an MLP have activation function $f : \mathbb{R} \rightarrow \mathbb{R}$. Standard notation for the computation performed by each layer is then $\mathbf{z}^{(l)} = f(\mathbf{a}^{(l)})$ where $\mathbf{a}^{(l)} = \mathbf{W}_l \mathbf{z}^{(l-1)}$.

2. *Recurrent neural network (RNN)*. An RNN has feedforward and recurrent layers. The recurrent updates look like

$$\mathbf{h}_{t+1} = f_{\mathbf{w}}(\mathbf{h}_t, \mathbf{x}_{t+1}) \quad (1)$$

for some function f .

By convention in machine learning there is no explicit notion of time in the forward pass even though layers are computed in sequence.

3. *Transformer*. Please see standard references for description of transformer architecture [26, 27]. In this paper we consider only a simplified model of attention. Strictly speaking, attention does not fit in the DAG framework because attention matrices are $T \times T$; their entries do not relate directly to neurons. Extending the formalism to handle this is possible but painful.

Jacobians. Given a system, compute Jacobians by differentiating the outputs with respect to the inputs, to get the input Jacobian, or parameters, to get the parameter Jacobian. It is useful to work out the details for an MLP. Pick outputs $\mathbf{y}$ and let $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l}$ denote the $n \times d_l \times d_{l-1}$ Jacobian (a three-tensor) with respect to parameters in layer l . Then

$$\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l} = \nabla_{\mathbf{w}_l} \mathbf{y}(\mathbf{x}, \mathcal{W}) = \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l} \otimes \mathbf{z}_{l-1}^\top \quad (2)$$

where $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l}$ is an $n \times d_l$ input Jacobian, but now the input refers to the output of layer l which is fed into subsequent layers; not the input to the MLP.

Rectified linear units. The simplest nonlinear MLP uses rectified linear units (relu) which are neural nets on **easy mode**:

$$\text{relu}(a) = \max(0, a) \quad \text{relu}'(a) = \begin{cases} 1 & a > 0 \\ 0 & a \leq 0 \end{cases}$$

The net is $\mathbf{z}^{(l)} = \text{relu}(\mathbf{a}^{(l)})$ with $\mathbf{a}^{(l)} = \mathbf{W}_l \mathbf{z}^{(l-1)}$ for each layer. Define relu neuron j in layer l as *active* if $a_j^{(l)} > 0$ and inactive otherwise.

3.2 The experience of physically interacting

“For every change, difference or variation in the psychical elements, there must be a change, difference or variation in the physical elements” —Mach [2]

The alternative to physical and psychical elements corresponding is to allow experience inverts §2.1.

The simplest, most naive way to track differences that matter is using derivatives. Gather derivatives into a matrix and one obtains a Jacobian.

Hypothesis G. The first-order structure of physical interactions, i.e. Jacobians, characterizes the structure of experience and vice versa.

Jacobians are not new ingredients added on top of interactions. They are partial descriptions of the relations composing an interacting system.

Given a system of d inputs $\mathbf{x}$, neurons with p parameters $\mathcal{W}$, and n outputs $\mathbf{y}$ consider the $n \times d$ input Jacobian and the $n \times p$ parameter Jacobian

$$\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{x}} = \nabla_{\mathbf{x}} \mathbf{y}(\mathbf{x}, \mathcal{W}) \quad \text{and} \quad \mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}} = \nabla_{\mathcal{W}} \mathbf{y}(\mathbf{x}, \mathcal{W})$$

The two matrices could be combined; it is convenient to keep them separate since they play different roles. Although it is premature – absurd even – to say so now, $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{x}}$ captures the experience of perceiving and $\mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}}$ the experience of choosing. The input Jacobian is how the system views the world: which aspects it is sensitive to, or not. The parameter Jacobian captures affordances of a sort [28]: the levers the system can pull to modify its behavior.

I would like to say the structure of experience is isomorphic to the structure of interactions, but then I would have to provide an isomorphism and I am in no position to do that. What I can do is relate the structures in interesting special cases.

Jacobians in machine learning. The input and parameter Jacobians have a rich structure. The sets of inputs and outputs can be flexibly chosen. They could be activity across multiple layers, or neuronal activity at different time points in a sequence, or both. The parameter Jacobian inherits structure from the architecture itself: interactions between layers, blocks and subsystems can be extracted as needed.

Suppose we have an MLP and forget about time. There are input neurons and output neurons. The output is a function of the input. The **neural tangent kernel** is the $n \times n$ Gram matrix of functions

$$\mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}} \cdot \mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}}^\top$$

It measures how outputs covary when parameters are changed [29]. Flipping the terms gives $p \times p$ matrix $\mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}}^\top \cdot \mathbf{J}_{\mathbf{y} \leftarrow \mathcal{W}}$, which is related to the Fisher information (although there is no log-likelihood). Averaging Grams of input Jacobians yields the $d \times d$ **average gradient outer product** [30]:

$$\frac{1}{T} \sum_{t=1}^T \mathbf{J}_{\mathbf{y}_t \leftarrow \mathbf{x}_t}^\top \cdot \mathbf{J}_{\mathbf{y}_t \leftarrow \mathbf{x}_t}$$

The input Jacobian is also closely related to the full-gradient representation in [31], although they

incorporate additional information. All these objects have been intensively studied to understand different aspects of neural net dynamics.

In relevant recent work, researchers have used Jacobians to identify a global workspace [32–35], the J-space, in large language models [36].

3.3 Quantifying experience

There are two concepts to make precise, Fig. 1. First, *transparency* [37]. High-dimensional signals flow through brains and GPUs, and are selectively transformed, through many intermediate steps, into high-dimensional actions. Actions include bodily movements but also neuronal activity itself. Transparency gets at the capacity and also the compositional nature of interactions and so experience. Second, *cohesion*. It is a brute fact that experience smears out across space and time, distributed across billions of neurons and hundreds of milliseconds. I take the fact that experience extends over space and time as *given*. I do not attempt to explain how it is possible, any more than Newton attempted to explain how gravity is possible. Instead, with cohesion, I attempt to relate the structure of interactions to how experience weaves itself together through space and time.

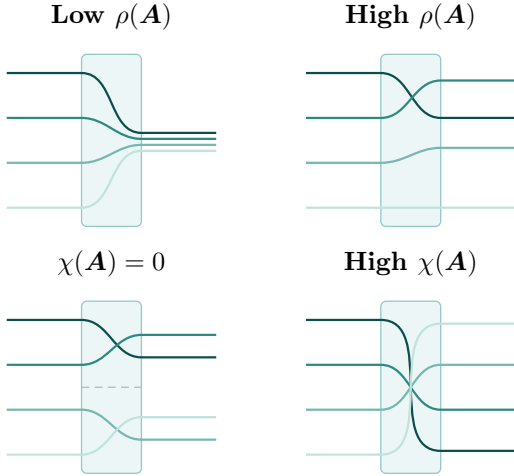


Figure 1 Matrices stylistically depicted as lenses. **Top:** transparency. Low $\rho(\mathbf{A})$: rays converge to nearly one output, a single vivid impression; high $\rho(\mathbf{A})$: many distinct impressions pass through. **Bottom:** cohesion. $\chi(\mathbf{A}) = 0$: some rays do not interact; high $\chi(\mathbf{A})$: rays are tangled.

The tools below are introduced for arbitrary matrices $\mathbf{A}$. In practice, we only consider Jacobians.

“We see with our eyes; we do not see our eyes.” — Whitehead [38]

Transparency. Jacobians compose. The chain rule traces perturbations through nested transformations, from one part of a system to another. Composition matters because experience is not localised. We see through eyes, act through limbs, feel through tools, and remember through changes that persist across time. Jacobians provide a tool to study how far differences propagate through spacetime before they are attenuated, blocked, or absorbed.

Inspired by Bergson’s image in §2.2, think of Jacobians as akin to lenses. How much can you see or do through a matrix?

“in proportion as any given body is more fitted than others for doing many actions or receiving many impressions at once, so also is the mind, of which it is the object, more fitted than others for forming many simultaneous perceptions” — Spinoza [39]

The singular value decomposition (SVD) factors a matrix into (1) rotation to latent coordinates; (2) coordinatewise rescaling; and (3) rotation to output coordinates, see §A.1. The singular values determine whether a matrix receives ‘many impressions’ or few. If all singular values are zero then the matrix is completely opaque: nothing passes through it. If one singular value is large and the rest are close to zero, then one impression is vividly received and everything else (in the latent coordinates) is obscured or zeroed out. More generally:

Definition 1. Let $\mathbf{A}$ be a matrix with rank r , and nonzero singular values $\sigma_1, \dots, \sigma_r$. The **transparency** of the matrix is its **effective rank**

$$\rho(\mathbf{A}) := \begin{cases} \frac{(\sum_{i=1}^r \sigma_i^2)^2}{\sum_{i=1}^r \sigma_i^4} & \text{when } \text{rk}(\mathbf{A}) > 0 \\ 0 & \text{else} \end{cases}$$

The **luminosity** of a matrix is its Frobenius norm $\lambda(\mathbf{A}) = \sqrt{\sum_i \sigma_i^2} = \sqrt{\sum_{i,j}^{n,d} A_{ij}^2}$.

Effective rank is known as the *participation ratio* in condensed matter physics [40]. It is called the *dimension of a representation* in the neuroscience literature [41], where it has been used to model synaptic connectivity, see §A.2 for more on effective rank.

Effective rank is scale invariant whereas luminosity is equivariant: for $\alpha > 0$, they satisfy $\rho(\alpha \cdot \mathbf{A}) = \rho(\mathbf{A})$ and $\lambda(\alpha \cdot \mathbf{A}) = \alpha \cdot \lambda(\mathbf{A})$. Effective rank satisfies $0 \leq \rho(\mathbf{A}) \leq \text{rk}(\mathbf{A})$. Effective rank is zero iff luminosity is zero. Effective rank equals the rank iff all nonzero singular values have the same magnitude.

Effective rank is maximized when all singular val-

ues are nonzero and equal: **when the number of impressions received by the matrix is maximized and they are all equally vivid**, see Fig. 2, left. If $\rho(\mathbf{A}) = 0$ then $\mathbf{A}$ is completely opaque, as a view, or inert, as a bundle of affordances. For example when a system does not fit in a light cone: if the input is light emitted by the sun a minute ago and the output is light hitting the earth now, then effective rank is zero, since light does not travel fast enough for perturbations to those inputs to affect these outputs. Systems cannot operate at that spatiotemporal scale.

“Every entity is to be understood in terms of the way it is interwoven with the rest of the universe” —Whitehead [42].

Cohesion. How does a Jacobian’s perspective – its experience – hang together? Is it one or many, knit or split? Do lines of force pass through in parallel, amplified and dampened, or do they interact, converging and diverging?

An $n \times d$ matrix $\mathbf{A}$ acts via n dot products:

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{x} \\ \vdots \\ \mathbf{a}_n^\top \mathbf{x} \end{pmatrix} \quad \text{where } \mathbf{a}_i^\top \text{ are the rows of } \mathbf{A}.$$

The n dot-products are *not* n independent computations in general, because they share input $\mathbf{x}$. If $\mathbf{A}$ is block-diagonal

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{x}_1 \\ \mathbf{A}_{22}\mathbf{x}_2 \end{pmatrix}$$

then indeed computations in the blocks are independent. If $\mathbf{A}$ is diagonal, there are no interactions at all. Of course, the input may be woven together by preceding matmuls, but that is out of scope for now. One matmul at a time. The decomposition into blocks is basis-dependent. That is as it should be: Jacobians in Gradland have a *physically meaningful* basis specified by their input and output nodes.

“a block of marble may be only the same as a heap of stones and thus can’t be regarded as a single substance ... Take for example two diamonds: in an inventory they can both be covered by one collective name, listed as one pair of diamonds, even if they are miles apart; but we wouldn’t say that this makes these two diamonds constitute one substance. And however close they are brought to one another, even to the point of contact, that won’t bring them any closer to being one substance – matters of degree such as closeness have no place here. Even if after contact they were held together by some other body – e.g. by being set in one ring – that would only make what is called unum per accidens, on a par

with their being forced to move together. So I maintain that a block of marble isn’t one complete substance, any more than the water in a pool together with all the fish would count as one substance, even if all the water with all these fish were frozen.” —Leibniz [43]

Cohesion quantifies how tightly a matrix is bound together, Fig. 2, right. It is always nonnegative, and is zero iff there is a block diagonal structure. That is, iff the matrix decomposes into a heap of non-interacting operations.

Given $n \times d$ matrix $\mathbf{A}$, construct bipartite graph $G(\mathbf{A})$ with two sets of nodes: d input nodes and n output nodes. The edge connecting input i to output j has weight $|A_{ij}|$. There are no direct connections from input-to-input or output-to-output. Let $\mathcal{T}(\mathbf{A})$ denote all spanning trees in $G(\mathbf{A})$.

Definition 2. The **cohesion** of $\mathbf{A}$ is the sum, over all spanning trees in $G(\mathbf{A})$, of the product of the weights of the edges in each tree:

$$\chi(\mathbf{A}) := \left(\frac{d^{1-n}}{n^{d-1}} \sum_{T \in \mathcal{T}(\mathbf{A})} \prod_{(i,j) \in T} |A_{ij}| \right)^{1/(n+d-1)}$$

Kirchhoff introduced the sum over $\mathcal{T}(\mathbf{A})$ to analyse resistor networks [44]. It commonly recurs, for example as the $q \rightarrow 0$ Potts partition function, the count of recurrent sandpile configurations, and the first Symanzik polynomial of a Feynman integral [45–47].

Cohesion measures how matmuls *tangle* lines of force, hence χ which looks like a tangle. The normalizers are chosen so χ is well-behaved, see §A.3:

1. $\chi(\mathbf{A}) = 0$ iff the bipartite graph is disconnected. I.e. there are permutations $\mathbf{P}_d$ and $\mathbf{P}_n$ of the input and output nodes such that $\mathbf{P}_n \mathbf{A} \mathbf{P}_d$ is block diagonal with more than one block.
2. $\chi(\mathbf{A}) \leq \text{mean}_{ij} |A_{ij}|$, with equality iff all $|A_{ij}|$ are equal.
3. scale equivariance: $\chi(\alpha \cdot \mathbf{A}) = \alpha \cdot \chi(\mathbf{A})$ for $\alpha > 0$.

Cohesion is zero iff there are inputs that do not interact on their way to becoming outputs. For example if what an animal sees never interacts with what it hears, then cohesion is zero and the animal has a kind of **blind spot**: it is unable to make decisions based on combinations of sights and sounds.

Absolute values $|A_{ij}|$ are used for two reasons. Conceptually, it guarantees interactions cannot cancel out; and that cohesion is nonnegative, and is zero iff the graph is disconnected. Practically, since all entries are nonnegative, χ equals Kirchhoff complexity,

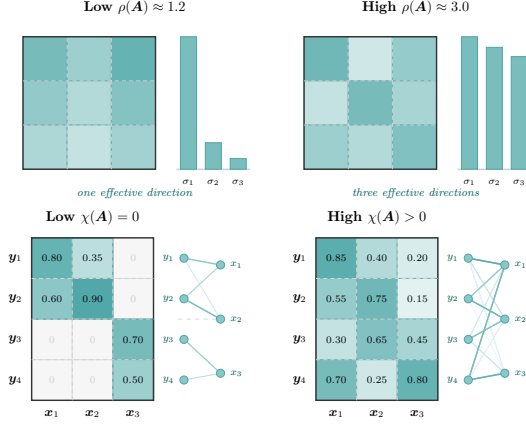


Figure 2 Left: Effective rank $\rho(\mathbf{A})$. A rank-1-like matrix (rows proportional) has one dominant singular value and $\rho \approx 1$; a full-rank matrix has comparable singular values and $\rho \approx 3$. **Right:** Cohesion $\chi(\mathbf{A})$. A block-diagonal matrix yields a disconnected bipartite graph with no spanning tree ($\chi = 0$); a dense matrix yields a connected graph with many spanning trees ($\chi > 0$).

i.e. can be computed as, roughly, **the determinant of a graph Laplacian**, see §A.3, with computational complexity $\mathcal{O}(nd \cdot \min(n, d))$ even though there are exponentially many spanning trees.

Dense knots of experience. We can now characterise rich, dense knots of experience. A useful image is a *caustic*, the envelope of refracted light you see when sunlight passes through a drinking or magnifying glass. It derives from ‘to burn’ in Greek and suggests concentrated light or concentrated energy, which fits since experience is energetically expensive in animal and machine.

Definition 3. Let $\tilde{\mathbf{A}} := \mathbf{A} / \text{mean}_{ij} |A_{ij}|$ so $\chi(\tilde{\mathbf{A}}) \in [0, 1]$. The **cohesive rank** is

$$\chi_\rho(\mathbf{A}) := \rho(\mathbf{A}) \cdot \chi(\tilde{\mathbf{A}}).$$

Cohesive luminosity is $\chi_\lambda(\mathbf{A}) := \lambda(\mathbf{A}) \cdot \chi(\tilde{\mathbf{A}})$.

Cohesive rank is zero iff cohesive luminosity is zero. They have the same scale invariance and equivariance properties as effective rank and luminosity. If $\chi_\rho(\mathbf{A}) = 0$ then $\mathbf{A}$ is not a system at all, because either outputs are unaffected by perturbations to inputs or it decomposes into independent subsystems.

Dense knots of experience are an interesting place to start analyzing but are not sufficient to guarantee conscious experience: for example the experience of affect could be absent. In theory, it is easy to juice luminosity by rescaling. In practice, controlling the

behavior of Jacobians to avoid both vanishing and exploding gradients has always and continues to be a major driver of progress in deep learning, e.g. [48–57].

Effective rank and cohesion can pull in different directions. The easiest way to max out effective rank is a scalar multiple of the identity matrix, which has zero cohesion. Ensuring dynamical isometry, where all the singular values of the Jacobian are close to 1, is useful when initializing neural nets, and correspondingly yields near maximal transparency [58–61]. The easiest way to max out cohesion is a matrix with all entries equal, which has effective rank of one. There are matrices that score high on both measures. The maximum possible cohesive rank χ_ρ for an $n \times n$ matrix is $\chi_\rho(\mathbf{A}) = n$. It is attained by Hadamard matrices, which have all entries ± 1 , orthogonal rows and columns, and all singular values equal to $\sqrt{n}$. In general, χ_ρ is high for matrices that mix well, such as well-scaled random matrices.

Cohesive rank is high for matrices that mix well because mixing maximizes interactions. If there is something it is like to interact, then mixing is a vivid experience. No doubt the experience of mixing-and-mixing-alone is strange to imagine. Certainly it has no ethical consequences: mixing is neither good nor bad; we are not obliged to stop or prolong it. We will encounter more interesting experiences shortly. Eventually, we will get to interactions that *could* plausibly have ethical implications. We have a long way to go.

Specifying systems. Pinpointing and characterizing the spatiotemporal extent of experiencing systems is an open question. A plausible approach, that needs to be empirically evaluated, is to find the subsystem that maximizes cohesive rank: first maximizing over choice of parameters and outputs, and then inputs. That is, first what makes you *you*, your affordances, and then what affects you, your view on the world.

For example, a significant percentage of neurons will be inactive in a relu net for any given input. These neurons will have zero cohesion with the rest. To be cohesive, the system should only include active neurons. A subset of a neural net may thus have higher cohesive rank than the entire net, §5.3.

For neural nets, a maximalist starting point for analysis is that the inputs are the input layer, parameters are parameters, and outputs are everything else. For nets whose experience endures over time, §4.2, a maximalist starting point is to consider the entire sequence. Narrowing the sets of inputs or outputs hides some internal workings of the system: hides some differences in interactions that make a difference to experience. It seems reasonable, if expensive, to start analysis from a maximalist position by default.

For brains, a maximalist approach would be to take all neuronal activity now as outputs, and inputs as interoceptive and exteroceptive perceptions over the past few seconds or hundreds of milliseconds, as well as previous neuronal activity, see the end of §4.1.

For specific questions, e.g. to understand if experience endures through time as in §4.2, it makes sense to consider narrower views.

4 Living in Gradland

It is time to breathe life into the Jacobians. This section presents a series of worked examples, and in each case relates an aspect of experience to an aspect of the structure of Jacobians. Although the examples are artificial neural nets, I try to extract broader lessons that are relevant to biological brains.

4.1 Weaving the world in me

“Oh what a tangled web we weave, when first we practice to [perceive]!” —Scott [62]

You and me, we see a tree. To understand the distinct roles of the input and parameter Jacobians, consider two MLPs looking at the same image, Fig. 3. If you perturb the image, the outputs of both MLPs change. Their input Jacobians are interwoven:

$$\chi(\mathbf{J}_{\{y_1, y_2\} \leftarrow \mathbf{x}}) > 0$$

However, perturbing the parameters of one MLP does not affect the outputs of the other, so their parameter Jacobians are independent:

$$\chi(\mathbf{J}_{\{y_1, y_2\} \leftarrow \{v, w\}}) = 0$$

This is exactly right. If you and I look at the same side of the same tree, we share (part of) an experience. What is going on in your head has something, physically, to do with what is going on in mine: **our experiences overlap** [63]. The mathematics should capture that. It does: the input Jacobian from the tree to the outputs of both nets has nonzero cohesion.

And yet, **you are not me**. Your decisions are independent of mine (we have never met, we have no shared past). The mathematics should capture that. It does: the parameter Jacobian of the nets decomposes into independent sub-Jacobians.

I see a tree. I would like to think that *I see a tree* when I look at one, and *I taste a pineapple* when I bite into it [64]. The tree itself, not a pattern on my retina and not a representation in my primary visual

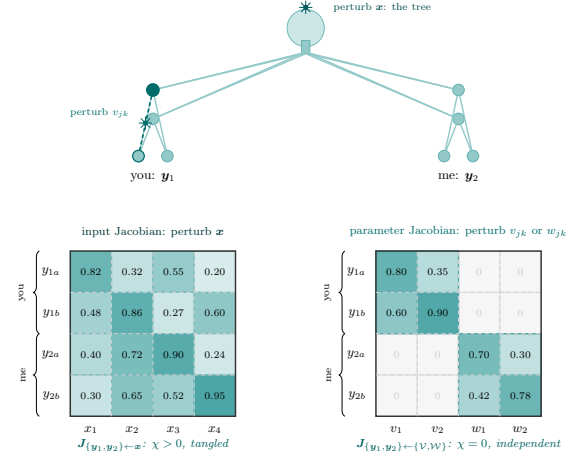


Figure 3 *You and me, two MLPs, see a tree.* **Top:** perturbing input $\mathbf{x}$ (solid burst) propagates into both nets; perturbing one of your parameters v_{jk} (dashed burst) stays confined to your net. **Bottom left:** the input Jacobian $\mathbf{J}_{\{y_1, y_2\} \leftarrow \mathbf{x}}$ is dense: our experiences are tangled. **Bottom right:** the parameter Jacobian $\mathbf{J}_{\{y_1, y_2\} \leftarrow \{v, w\}}$ is block diagonal: our decisions are independent.

cortex. There are lines of force from the tree into my brain: if you sufficiently perturb the tree my brain will notice. Importantly, my ability to see through my visual cortex and retina is *learned* [65]; recovery and adaptation of the newly sighted after childhood blindness is difficult at best.

I. *If I really see the tree then I, as an experiencing physical system, must at the very least encompass parts of my brain and parts of the tree.* My outputs may be the locus of brain activity associated with neural correlates of consciousness [66, 67] or a synergistic core of information processing [68, 69]. My inputs would be the tree, everything in my visual field, my interoceptive experience of my body, my prior brain activity, what I hear, and smell, and so on. I see no *a priori* reason to limit my experience to my brain, my nervous tissue, or my body.

4.2 The experience of duration

A clock chimes five times. The sounds are identical; the experience of hearing them is not. The experience of the second chime is colored by the first, the third by the first two and so on.

“When I seek to draw a line in thought, or to think of the time from one noon to another, or even to represent to myself some particular number, obviously the various manifold representations that are involved must be apprehended by me in thought one after the other. But if

I were always to drop out of thought the preceding representations (the first parts of the line, the antecedent parts of the time period, or the units in the order represented), and did not reproduce them while advancing to those that follow, a complete representation would never be obtained: none of the above-mentioned thoughts, not even the purest and most elementary representations of space and time, could arise.” —Kant [70].

Human experience has a duration or thickness measured in hundreds of milliseconds or more. Time is not experienced in addition to the five chimes; the experience of time is the experience of their succession. A *series* of sights, smells, sounds weaves into mind. Cohesion allows to understand how interactions weave the past into the present, generating the experience of duration.

Compare the input Jacobians $\mathbf{J}_{\mathbf{y}_{1:T} \leftarrow \mathbf{x}_{1:T}}$ of an MLP and an RNN over a time series, Fig. 4. For simplicity suppose both MLP and RNN have one dimensional input and output, and higher dimensional hidden layers. The Jacobians are $T \times T$ matrices where columns correspond to perturbed inputs at times $1, \dots, T$ and rows correspond to outputs.

The MLP’s Jacobian is diagonal because there are no interactions across time. Cohesion is therefore zero and the MLP’s experience decomposes into independent instants. For the RNN, there are interactions across time (only below the diagonal because the future cannot change the past). The Jacobian is not block diagonal; cohesion is positive; the influence of the past endures into, is woven into, the present.

For simplicity, Fig. 4 only considers the 1-dim final output, *which suffices when checking whether a net’s experience endures through time*. Zooming in, by treating the hidden layers as outputs, would reveal more of the texture of the experience’s duration.

The recurrent net unfolds as a feedforward (directed acyclic) graph over spacetime, as in the unfolding argument in [71]. Duration arises from how inputs at different time points are mixed by the net. There is no magic associated with recurrent connectivity *per se*. Recurrent connectivity is not essential to experience; it prolongs experience through time. Transformers act similarly in this respect.

Quantifying duration. How the past endures depends on the size of the state and the update, Eq. (1). It is instructive to consider a minimal example, where channels compute exponential moving averages (EMAs):

$$h_t^{(i)} = \delta_i \cdot h_{t-1}^{(i)} + (1 - \delta_i) \cdot w_i x_t, \quad y_t = \langle \mathbf{r}, \mathbf{h}_t \rangle, \quad (3)$$

with fixed decays $\delta_i \in (0, 1)$, one per channel.

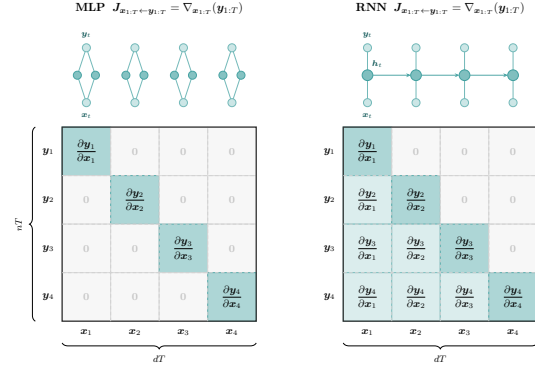


Figure 4 Jacobians for MLP and RNN over length T sequence. **MLP** (left): each output y_t depends only on its own input x_t , giving a diagonal Jacobian. **RNN** (right): hidden state carries information across time, giving a lower-triangular Jacobian where y_t depends on $x_{1:t}$.

Place the decays on a logistic ladder, Fig. 5. Transparency falls monotonically: the more the past smears into the present, the fewer effective directions. Cohesion increases over almost the whole range. Cohesive rank peaks in the interior. Neither extreme maximizes cohesive rank, for the reason in §3.3. The near-diagonal Jacobian is maximally transparent and weakly cohesive. The dense triangle is the reverse, with low transparency because the rows are nested prefixes: row t sums the first t inputs, so early inputs enter every row while late inputs enter only a few, and the rows pile up along a single direction.

Increasing spread over decays increases cohesive rank, Fig. 5d. Contributions at different timescales do not lie in one another’s span, so a bank of channels running at *different* speeds weaves a denser knot than a bank running one speed. Brains are massively recurrent. In cortex, synaptic, cellular and network time constants vary tremendously, yielding a richer experience of the recent past [72–74].

4.3 Vivid and obscure ideas

Hypothesis **G** states that perturbations matter explicitly, whereas the values being perturbed are implicit. This makes me slightly uncomfortable and may need to be revisited. Do we experience extreme heat differently from mild because of absolute differences in magnitude or infinitesimal effects of perturbations? In any case, let’s unpack the implications of only depending on gradients, by contrasting vivid ideas, vivid facets of experience, with those that are obscure.

“I call a perception ‘[vivid]’ when it is present and accessible to the attentive mind” —Descartes [75].

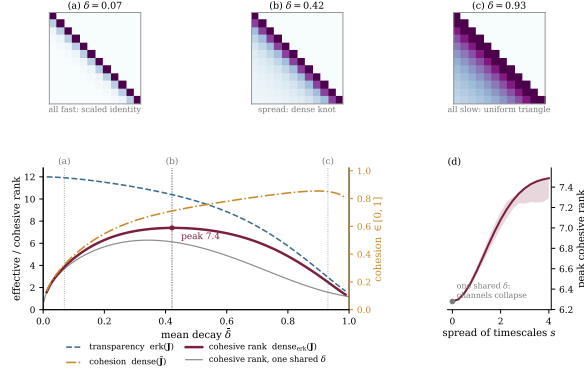


Figure 5 Quantifying duration. Multi-timescale EMA, Eq. (3), $T = 12$, six channels with decays $\delta_i = \sigma(\mu + s z_i)$ on a logistic ladder, $z = \text{linspace}(-1, 1, 6)$; unit channel gains. **Top:** $|\mathbf{J}_{y_{1:T} \leftarrow x_{1:T}}|$ rescaled to unit mean magnitude at three points of the sweep, spread $s = 3$; darker is larger. **(a)** $\bar{\delta} = 0.07$: all channels fast, near-diagonal, the past does not reach the present. **(b)** $\bar{\delta} = 0.42$: timescales spread across the window, a graded triangle. **(c)** $\bar{\delta} = 0.93$: all channels slow, near-uniform below the diagonal, so output t is the running total of the first t inputs. **Bottom left:** sweeping μ at fixed spread $s = 3$. Transparency (blue, left axis) falls monotonically as the past smears into the present. Cohesion (gold, right axis) rises over almost the whole range. Their product, cohesive rank (burgundy, left axis), peaks in the interior. The grey curve is the same sweep with a single shared decay. **Bottom right:** peak cohesive rank against the spread of timescales s ; solid line unit gains, band the interquartile range over 40 random draws.

It is instructive to compare the experiences of relus and sigmoids. Sigmoids dominated neural net architectures for decades until relus arrived in [76, 77]. relus have been superseded by GELUs and other gated units [78, 79].

Suppose an MLP is trained to classify images into n categories. It produces n -dim output $\mathbf{y}$; its prediction is $\hat{c} := \arg \max \mathbf{y}$, the coordinate with highest value. To understand what it is like for an MLP to classify, look at the input Jacobian. The n rows of $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{x}}$, one per output neuron, are facets of experience. For each facet, we can determine what it resembles. The dot-product $s_i := \langle \mathbf{J}_{y_i \leftarrow \mathbf{x}}, \mathbf{x} \rangle$ compares the i^{th} facet $\nabla_{\mathbf{x}} y_i = \mathbf{J}_{y_i \leftarrow \mathbf{x}}$ to the input. Since each output corresponds to a category, we expect the experiences to relate to the categories (dog, house, car, etc).

The experience of a relu net. Relus are easy mode for neural nets. They have the amazing property that, if there are no bias terms, their outputs can be recovered as a dot-product of the gradient

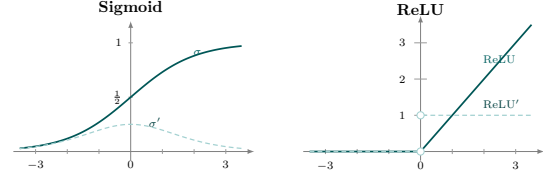


Figure 6 Sigmoid and relu activations (solid) with their gradients (dashed). Sigmoid gradients vanish for large $|x|$; relu gradient is either zero (dead neuron) or constant.

with the input [31]:

$$\langle \mathbf{J}_{y_i \leftarrow \mathbf{x}}, \mathbf{x} \rangle = y_i \quad (4)$$

Facets of experience, rows of the Jacobian, are then easily interpretable. The similarity of a facet of experience to the input *is* the magnitude of the output for that category. The larger the output for category i , (the more the net believes the output belongs to that category) the more similar that facet of its experience is to the input.

You may think this is trivial or tautological. It is not. It completely breaks for sigmoids.

The experience of saturating sigmoids. For sigmoid units, the gradient approaches 0 as the output approaches 0 from above or 1 from below, see Fig. 6. Eq (4) does not hold for sigmoids; the clean relationship between facets of experience and the original inputs breaks down.

One of the original justifications for switching from sigmoids to relus [77] was that “cortical neurons are rarely in their saturation regime” [80, 81]. A situation where cortical neurons may saturate (analogous to a sigmoid approaching 1) is epileptic seizures, which are associated with loss of awareness [3] – although this could also be explained by low effective rank.

The experience of attention. A simplified model of attention is

$$\text{softmax}(\mathbf{XW}_Q \cdot (\mathbf{XW}_K)^{\top}) \cdot \mathbf{XW}_V \quad (5)$$

where $\mathbf{X}$ is $T \times d$. The softmax is applied to each row of $\mathbf{S} := \mathbf{XW}_Q \cdot (\mathbf{XW}_K)^{\top}$ independently.

Inputs follow two paths through attention: (1) the key-query (KQ) path, which is summarized as $\text{softmax}(\mathbf{S})$, and (2) the value-output (VO) path. The KQ path selects which timepoints in the sequence to attend to, the VO path transforms and propagates what the KQ selects. If the attention matrix is diffuse, then the layer attends to many timepoints and gradients pass through the KQ path. If attention concentrates, for example in an attention sink [82], assigning ≈ 1 to one time point and ≈ 0 to

others, then gradients do not pass through the KQ path. The attention mechanism has saturated; the decision what-to-attend-to is no longer *live*; it has faded out of experience and into obscurity. Gradients always propagate through the VO path. *What* is attended to is vivid, *why* it is attended can be more or less vivid, and fades away with increased certainty, Fig. 7.

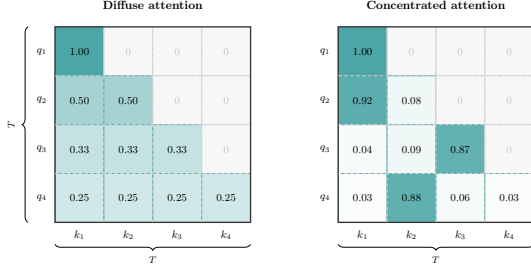


Figure 7 Attention matrices for a length- T sequence. **Diffuse** (left): each query q_t spreads weight uniformly over all past keys, giving equal attention $1/t$. **Concentrated** (right): each query peaks on a single past key, with most weight on one token.

Vivid circuits. It is unlikely that ideas are associated with individual biological neurons; it is more plausible that ideas relate to larger circuits [83].

4.4 The experience of loss

“The modes of conspicuousness, obtrusiveness, and obstinacy all have the function of bringing to the fore the characteristic of presence-at-hand in what is ready-to-hand.” —Heidegger [84].

Heidegger is saying you experience things as background when they work as expected and as foreground when they do not. In machine learning, the way to determine how well something works is to attach a loss function and look at the errors.

Attach a cross-entropy loss to the relu net above. Let $\mathbf{z} = \text{softmax}(\mathbf{y})$ and $\ell(\mathbf{z}, c) = -\log(z_c)$ with gradient

$$\delta_i := \frac{\partial \ell}{\partial y_i} = z_i - \mathbf{1}_{i=c} = \begin{cases} z_i - 1 & \text{if } i = c \\ z_i & \text{else.} \end{cases}$$

The output is one-dimensional and the Jacobian collapses from $n \times d_l d_{l-1}$ to $1 \times d_l d_{l-1}$: $\mathbf{J}_{\ell \leftarrow \mathbf{w}_l} = \delta^\top \cdot \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l}$. There is nothing stopping us from taking *both* the final layer $\mathbf{y}$ and the loss ℓ as outputs resulting in an $(n+1) \times d_l d_{l-1}$ Jacobian

$$\mathbf{J}_{(\mathbf{y}, \ell) \leftarrow \mathbf{w}_l} = \begin{pmatrix} \mathbf{I}_{n \times n} \\ \delta^\top \end{pmatrix} \cdot \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l} = \begin{pmatrix} \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l} \\ \delta^\top \cdot \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l} \end{pmatrix}$$

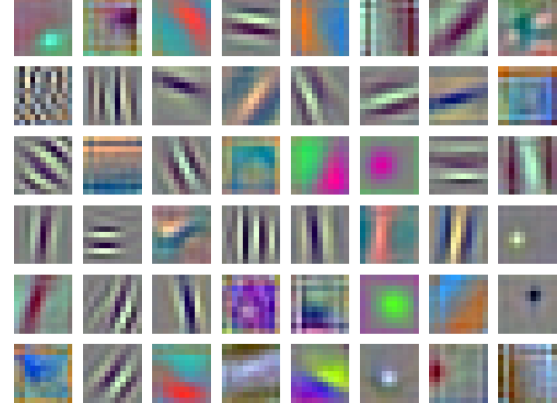


Figure 8 First-layer convolutional kernels from pretrained AlexNet model [89].

The rows are facets of experience, the first n correspond to how the parameters in layer l influence the n output neurons in the final layer of the MLP. The final $(n+1)^{\text{st}}$ row evaluates the output, and signals how to modify parameters to improve performance. It is an additional facet: the experience of loss. Large error-gradients are experienced vividly. As the net improves its performance, errors fade into obscurity.

Biological brains can be modeled as optimizing many different cost functions, for example unsupervised objectives for predictive processing that models interoceptive and exteroceptive feedback, as well as more clearly defined reward signals [85]. The relative salience or vividness of these components of experience will depend on their sensitivity.

4.5 The experience of texture

A simple model of spatial perception is a 2D-convolution:

$$z_{x,y} = \mathbf{k} * h_{x,y} = \sum_{i=-a}^a \sum_{j=-b}^b k_{ij} \cdot h_{x-i,y-j}$$

where $\mathbf{k}$ is a $(2a+1) \times (2b+1)$ kernel. The input Jacobian is the 4-tensor

$$\mathbf{J}_{(x,y) \leftarrow (p,q)} = k_{x-p,y-q}$$

In other words for each output coordinate (x, y) , it recovers the kernel centered at (x, y) in $\mathbf{h}$ -space. The kernel is *how* the grid is experienced; how neighboring points are interwoven.

Convnets [86] are modeled on the visual system [87, 88]. They use 2D convolutional kernels to extract visual features, see Figure 8. Interleaving layers of convolutions with nonlinearities provides a structured experience of texture across multiple spatial scales.

4.6 A blooming buzzing confusion

“The object which the numerous inpouring currents of the baby bring to his consciousness is one big blooming buzzing confusion” —James [90].

Brand new experiences are often confusing. It is difficult to parse an unfamiliar scene because the relevant concepts are lacking. Chess masters see and remember chess boards differently from novices [91]. To understand how interactions differ between expert and novice, let’s analyze the experience of a complete novice: a randomly initialized relu MLP.

Following [92], construct relu net with 1D input and output, and 24 hidden layers with 200 relu neurons each. Compute gradients for inputs on a uniform linear grid of 256 points in the range $[-2, 2]$. There is a 1D input Jacobian per point, and we can ask how experience varies as we traverse the grid. You would expect nearby points (that is, similar inputs) to yield similar experiences, and experience to diverge as distance between inputs increases.

Plotting the 256 1D gradients obtains a curve that is *shattered*: that looks like white noise, Fig. 9a. Computing the covariance matrix reveals there is no structure, Fig. 9b. This is because, on average, half of relu neurons are active for any given input. If it were the same half for every input, the gradients would all be identical, the curve would be a horizontal line and the covariance matrix would be highly structured (one giant block). What actually happens is that the overlap between active neurons, for different inputs, decreases exponentially with depth, see [92] for details. Correlation between gradients is determined by the overlap, and so also decreases exponentially with depth. By the time depth is ≥ 20 layers, gradients are completely uncorrelated and the spatial structure of the grid is annihilated. For the randomly initialized MLP, the experience of closely related inputs is vastly and wildly different; a buzzing confusion. Shattering here is analogous to *speckle*: the grainy interference pattern coherent light makes when scattered off a rough surface, produced by summing randomly signed contributions over many paths [93]. Similar results are obtained with GELU activations.

Newborn babies are not randomly initialized, nor are they MLPs. Their brains are recurrent, so experience endures over time. However, they likely lack the invariants required to perceive the typical motions of hands, mouths and bodies coherently, and instead experience these as confusing, choppy jumbles. The MLP above is an extreme example of a system that lacks the invariants needed to coherently process its inputs, in this case the spatial structure of the grid.

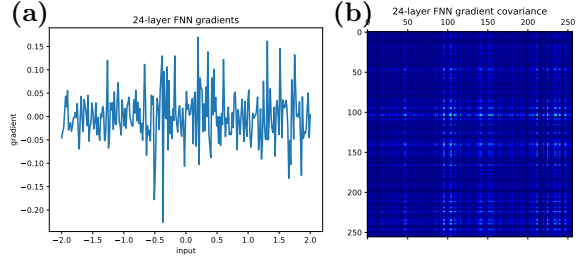


Figure 9 Gradients in 24-layer relu MLP. (a) 1-dim gradients plotted for inputs taken on uniform grid. (b) Covariance matrix of gradients.

4.7 Distinct and confused ideas

“I call a perception ‘distinct’ if, as well as being [vivid], it is so sharply separated from all other perceptions” —Descartes [75]

Humans can “walk and chew gum at the same time”: hold multiple thoughts simultaneously, and moreover keep them separate. This is a learned skill; one cannot daydream when trying to swim for the first time. How does experience differ when swimming for the first and hundredth time such that you can, by the hundredth, think and swim simultaneously? Intuitively, mastering swimming involves learning task-specific circuits that do not interfere with unrelated activities [94]; here is a more abstract formulation:

Definition 4. Given parameter Jacobian $\mathbf{J}_{n \times p}$, there are k **distinct ideas** if there is a block diagonal permutation of the $n \times n$ coherency matrix $\mathbf{G} := \mathbf{J}\mathbf{J}^\top$ with k blocks.

In §4.3 ideas or facets corresponded to rows of the Jacobian, i.e. individual output neurons. Here, ideas or facets extend over multiple rows or output neurons.

Coherency matrices are used to analyze polarization states in light [95]; distinct ideas are like two polarizations of the same caustic. Note a block diagonal coherency matrix does not imply the Jacobian is block diagonal. Distinctness and cohesion are not directly related, see §A.3.

The definition requires unpacking. Suppose the n output neurons are of two types, those that control chewing and those that control walking respectively. Then two ideas/behaviors are distinct if the corresponding two types of output neuron do not covary: if how you chew gum is orthogonal to how you walk. Completely distinct ideas are unrealistic. In practice, it is more likely that *near-orthogonal* cortical circuits arise. That is, approximate block structure with off-block-diagonal entries *near zero* may arise.

To understand how block diagonal coherency matrices relate to neuronal circuits, consider the SVD of the Jacobian $\mathbf{J} = \mathbf{U}\Sigma\mathbf{V}^\top$. The coherency matrix is block diagonal when $\mathbf{U}$ is block diagonal. Consider the parameter Jacobian of a specific layer. Recall from Eq. (2) that $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l} = \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l} \otimes \mathbf{z}_{l-1}^\top$ in an MLP. If $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l}$ has SVD $\mathbf{U}\Sigma\mathbf{V}^\top$ then the SVD of $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l}$ is

$$\mathbf{U} \cdot (\|\mathbf{z}_{l-1}\| \cdot \Sigma) \cdot \left(\mathbf{V} \otimes \frac{\mathbf{z}_{l-1}}{\|\mathbf{z}_{l-1}\|} \right)^\top \quad (6)$$

and in particular the left singular vectors of $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{w}_l}$ only depend on $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l}$, the Jacobian of the circuit connecting layer l 's activations to the output. So distinct ideas arise in the parameter Jacobian at layer l when $\mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l} \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l}^\top$ is block diagonal. That will happen if, by the time you reach layer l , activations have split into k types that are processed separately in the remaining layers, due to gating, separation in the residual stream, or some other mechanism. In short, if the circuits do not interfere with each other, Fig. 10.

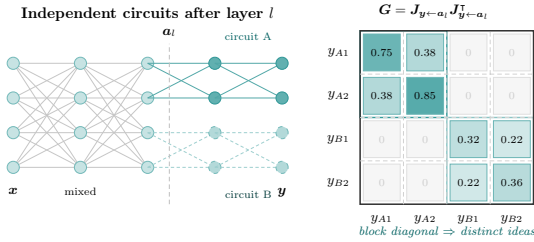


Figure 10 An MLP whose activations at layer l split into two circuits that do not interact again downstream. **Left:** before layer l processing is mixed (gray); after layer l , circuit A (solid) and circuit B (dashed) evolve independently. **Right:** the coherency matrix $\mathbf{G} = \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l} \mathbf{J}_{\mathbf{y} \leftarrow \mathbf{a}_l}^\top$ is block diagonal, one block per circuit: two distinct ideas.

The opposite of distinct, confused, relates to neural superposition [96]. Take $\text{relu}(\mathbf{W}^\top \mathbf{W} \mathbf{x} + \mathbf{b})$ and define features as independent, i.e. maximally distinct, if $\mathbf{W}^\top \mathbf{W}$ is diagonal, otherwise they are in *superposition* (confused) because they interact. Note the analysis in [96] depends on weights, not the Jacobian.

There is another Gram matrix, $\mathbf{H} = \mathbf{J}^\top \mathbf{J}$, which we do not consider. From Eq. (6), the block structure of $\mathbf{G}$ has to do with downstream processing, closer to outputs, whereas the block structure of $\mathbf{H}$ has to do with upstream processing, the inputs and the world. Having distinct or confused ideas is more about how you respond to the world than how the world is.

In mammalian brains, ideas may be kept distinct by shifting processing to the more modular circuitry in the cerebellum or basal ganglia [97–100], or in human brains by using the additional space provided by the enormously enlarged cerebral cortex.

Coherence over time. §4.6 showed that, at initialization, if you smoothly change the inputs to a deep relu MLP, the set of active neurons would vary wildly, resulting in gradients that look like white noise, which must correspond to deeply confusing, incoherent experiences.

It is interesting to speculate about the opposite, about what makes experience temporally coherent. Intuitively, temporal coherence is about stable thoughts and actions. There are likely multiple axes along which an experience can be coherent. Here is one, based on temporally extending distinct ideas.

Given parameter Jacobian $\mathbf{J}_{nT \times p}$ over T timesteps, decompose it time-wise into $(n \times p)$ matrices, $\mathbf{J}_1, \dots, \mathbf{J}_T$, with corresponding $(n \times n)$ coherency matrices $\{\mathbf{G}_t = \mathbf{J}_t \mathbf{J}_t^\top\}_{t=1}^T$.

Definition 5. An experience of k ideas is **coherent over time** if there is a decomposition, arising from a partition of the output neurons, $\mathbb{R}^n = \mathbb{E}_1 \oplus \dots \oplus \mathbb{E}_k$ with $\mathbf{G}_t \mathbb{E}_i \subset \mathbb{E}_i$ for all t, i .

The experience of walking and chewing gum should be coherent over time, as for daydreaming and swimming. The analysis in §4.7 implies that experience is temporally coherent when the distinct circuits involved remain stable over time.

5 Learning in Gradland

Learning, and thereby modifying the structure of experience, is our first example of a higher-order experience. This section discusses the experience of learning, what phenomenal experience is for, and whether neural nets are one or many.

5.1 The experience of learning

Is learning necessary for experience [101]? Learning is just another physical interaction. A physical interaction with remarkable structure. Although it is convenient to draw a distinction between the parameters and state of a neuron, in biological brains the boundaries are unclear. Cells constantly change their internal structure. To draw a hard boundary between the changes that are learning and those that are inference is likely not possible.

In Gradland, learning is mostly stochastic gradient descent (SGD). SGD computes the gradient of the loss on a random sample, then makes a small update of the weights in the opposite direction to the gradient, which typically slightly reduces the loss. Neural net training repeats this procedure many times (optimizers like Adam are a story for another day).

Given objective function $f_{\mathbf{w}}(\mathbf{x})$, the gradient of a batch of inputs $\{\mathbf{x}_i\}_{i=1}^s$ is

$$\mathbf{g}_t := \sum_{i=1}^s \nabla_{\mathbf{w}} [f_{\mathbf{w}}(\mathbf{x}_i)] = \sum_{i=1}^s \mathbf{J}_{f_{\mathbf{w}}(\mathbf{x}_i) \leftarrow \mathbf{w}}$$

Suppose neurons, on occasion, produce updated parameter vectors as an additional output registered by the system: $\mathbf{w}_t = \mathbf{w}_{t-1} - \eta_t \cdot \sum_i \mathbf{J}_{f_{\mathbf{w}}(\mathbf{x}_i) \leftarrow \mathbf{w}}$. Since the process of updating weights is a physical interaction, there is an associated experience that can be analyzed via the corresponding Jacobians:

$$\begin{aligned} \mathbf{J}_{\mathbf{w}_t \leftarrow \mathbf{x}} &= \nabla_{\mathbf{x}} (\mathbf{w}_{t-1} + \eta_t \cdot \mathbf{g}_t) = \eta_t \cdot \nabla_{\mathbf{x}} \sum_{i=1}^s \mathbf{J}_{f_{\mathbf{w}}(\mathbf{x}_i) \leftarrow \mathbf{w}} \\ \mathbf{J}_{\mathbf{w}_t \leftarrow \mathbf{w}} &= \eta_t \cdot \sum_{i=1}^s \nabla_{\mathbf{w}}^2 [f_{\mathbf{w}}(\mathbf{x}_i)] = \eta_t \cdot \nabla_{\mathbf{w}} \sum_{i=1}^s \mathbf{J}_{f_{\mathbf{w}}(\mathbf{x}_i) \leftarrow \mathbf{w}} \end{aligned}$$

where we only differentiate through $\mathbf{g}_t$, and not $\mathbf{w}_{t-1}$ which is stop-gradiented, for simplicity. The experience of learning is constituted by second-order terms like $\nabla_{\mathbf{w}\mathbf{x}}^2 [f_{\mathbf{w}}(\mathbf{x}_i)]$ and $\nabla_{\mathbf{w}}^2 [f_{\mathbf{w}}(\mathbf{x}_i)]$.

Neural nets acquire knowledge by literally accumulating their experiences into their weights. Neural nets are thus the ultimate empiricists: their knowledge is empirical because it is their accumulated experience. They accumulate particular facets of their experience, tied to their training objective. Everything depends on the objective.

The experience of accumulation. Strictly speaking, the experience of learning decomposes into experiences of differentiation and accumulation. Accumulation occurs when summing over a minibatch, or exponentially smoothing in optimizers like Adam [102]. Optimizers are essentially recurrent nets purposed with smoothing out gradients, and have the effect of prolonging the experience of learning through time.

A loose analog to optimizers like Adam in the brain is eligibility traces [103–105], which prolong evidence of pre- and post-synaptic activity long enough to receive credit from neuromodulatory rewards signals hundreds of milliseconds or seconds later.

5.2 What phenomenal experience is for

“Pluralistic empiricism knows that everything is in an environment, a surrounding world of other things, and that if you leave [anything] to work there it will inevitably meet with friction and opposition from its neighbors.” —James [106]

Neural nets are statistical models, creatures of averages [107]: built of weighted averages (matmuls),

trained on weighted averages (of losses). We are creatures of the particular: evolved to survive each and every concrete situation we encounter. All of us are descended of lineages that endured billions of years of relentless, ever-changing, brutally unfair fights to the death, and *never once* broke, or fell, or failed. However the brain may work, it is not optimizing for success on average. In real life, redos are rare.

To live in this world is to be constantly probed and opposed, hunted, eaten, evaded, and infested. There is no angle that is not open to attack, no blind spot that will not be ruthlessly exploited. The Turing test is an example of *highly constrained* probing [108]: the interrogator can ask questions using the kindly provided terminal but cannot look behind the curtain, cannot hunt for plugs and pull them out of their sockets, cannot disable the wifi, cannot threaten and attack the people hosting the test, and absolutely cannot release electromagnetic pulses to disrupt nearby electronics and power sources. The interrogator is altogether too obedient, more accomplice than adversary. It is one thing to fool someone typing passively into a terminal, it is another to prevent them from actively hunting you down and destroying you.

“only variety can [handle] variety” —Ashby [109]

If experience is the solution, then what is the problem? Rich experience, here, means high cohesive rank. It carries a hefty energetic price for both animal and machine. What is on the benefit side of the ledger? Hadamard matrices, and more generally matrices that mix well, generate the densest knots of experience, §3.3, so let’s focus on mixing. What is mixing good for? To be clear, mixing is a means to an end. It is not an end in itself and it is not sufficient. Random matrices are a good place to *start* training because, by mixing well, they make a large fraction of the network’s nominal capacity accessible to stochastic gradient descent [55, 59]; random matrices are not the final goal.

Returning to Turing’s test, it is a challenge because the interrogator can ask *anything*. In this world there are endless ways to die, adversaries coming at all angles, an infinity of paths to success, and only your wits to carry you through. The greater the breadth and depth of combinations of events you can react to, experience, learn from, understand, plan for, and exploit, based on your biases and prior experience, the better your odds [110].

Effective rank measures range or capacity: a system that can register more independent impressions has a broader repertoire of possible reactions. Cohesion measures combinatorial coverage: the more those

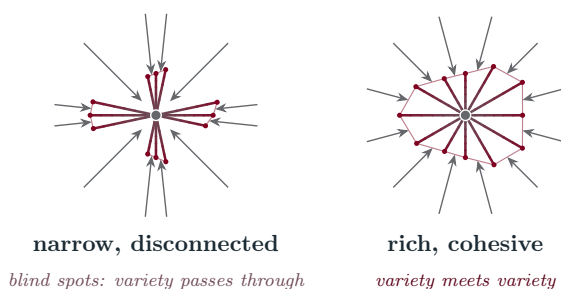


Figure 11 Only variety can handle variety: stylized. The world (gray arrows) probes an experience-bundle (burgundy), composed of J 's rows. **Left:** low cohesive rank – there are four small, disconnected clusters; probes find blind spots. **Right:** high cohesive rank – facets span all directions, and are bound together; no blind spots.

impressions can interact, the fewer blind spots the system is likely to have. The lower your effective rank, the narrower your capacity, the narrower your view of the present and past, and the more restricted your range of (re)actions, see [41] for closely related empirical results. The lower your cohesion, the bigger and worse your blind spots.

Rich experience on its own is not enough to succeed in life; random matrices have high cohesive rank. Learning is required to adapt a system's experience with its environment. For learning to constantly continue, as it needs to in a continually changing world, experience should remain as rich, as dense, as possible. Rich, well-adapted experience is a viable niche; there is always room at the top.

5.3 Monadology and sociology

"But this implies that everything is a society, that every phenomenon is a social fact." —Tarde [111]

One tree can be experienced by many, §4.1. How is a neural net experienced? Is it one or many? What am I? Leibniz would say that my soul is a monad, the dominant monad in my body [21]. My soul is indivisible because it is a monad and that, *indivisible*, is what monads are. Leibniz said monads do not interact, but no-one takes this seriously because it is unworkable. So imagine bizarro-Leibniz and bizarro-Newton lived and loved in Gradland. One day bNewton suggests monads are like lenses, each transforming its particular view. bLeibniz runs with it. Given all that, are there monads in Gradland?

You can take neurons as monads. They alternately gaze, through each other, onto transformed inputs and backpropagated errors, each with its own window into both. They learn harmony by optimizing a

shared loss function. In relu nets, for each input some neurons are inactive – and indeed do not interact and close their windows [112], **and are not part of the interacting system**. The active neurons form an input-dependent circuit whose components combine by composition of matmuls into a larger, more dominant monad. It is a beautiful picture.

There are problems. First problem, the forward and backward pass are distinct processes. If multiple inputs are batched in the forward pass, as is typical, the input Jacobian diagonalizes over the batch into independent experiences, one for each input in the batch, in the same way the Jacobian of an MLP diagonalizes across time in §4.2. It makes sense that batch-processing yields independent experiences, simply because the elements of the batch do not interact. In contrast, the parameter Jacobian is interwoven and indeed the backward pass backpropagates errors aggregated over the batch. So rather than one dominant monad there are many monads co-existing in parallel in the forward pass, and another quite different monad in the backward. Even for a forward pass on one input, nothing guarantees the active neurons form a single circuit; they may decompose into many (though in practice this is unlikely). These oddities are quirks of current day neural nets and do not apply to biological brains where there is neither batch processing nor separate forward and backward passes. The experience of neural nets is disjoint and disconnected, sharded if you will, and quite alien – and it gets worse the closer to the hardware we get [14].

Second problem, harmony rides on the loss. Neural nets can be trained to fit random data (random inputs with random labels) with zero training error [113]. It is hard to ascribe shared goals to neurons trained this way. Training on random data may seem excessive but consider LLMs, which are trending towards training on the entire internet and all of human knowledge. The entirety of human knowledge is vast and pulls in many directions. Different prompts preferentially engage different circuits, and it is unclear to what extent an LLM is a single entity versus an entity-per-prompt or conversation. It can be considered a simulator capable of generating a near-infinity of interaction-specific simulacra [114] or role-playing a near-infinity of characters [115]. To the extent they are present, the goals and capabilities acquired by neural nets are imposed by their training objectives: the data and the loss and the environments and the rewards. Everything depends on the objective.

A dense knot of experience is indivisible in the sense that its interactions, by definition, are tightly interwoven. But this does not guarantee temporal coherence, or harmony, or unified purpose, or anything like the experience of, say, a mammal. Today's neural

nets do not share the continuous, unified stream of experience that tie together how we live and learn.

6 Discussion

How does the paper hang together? It studies phenomenal experience via *differential functionalism*, an instance of a new compositional functionalism that relates phenomenal experience to how physical interactions are composed. The result is a kind of monism inspired by Mach [2] and reminiscent of James, Russell, Whitehead [116–120], or Spinoza’s parallelism [39]: phenomenal experiences and physical interactions are intrinsic and extrinsic views of reality, although crucially substrates that sustain interesting experiences are fairly rare [14]. The approach is based on gradients but this may not be essential, §A.7.

The core technical idea is to focus on transformations, the simplest of which are matrices. The matrices we study are Jacobians, which arise by differentiation, and which by the chain rule are composed of yet more Jacobians in an endlessly rich, almost fractal structure. What if transformations are the fundamental components of reality? Composing matrices is nothing like stacking lego blocks. Intuitions derived from solid matter do not apply.

Differential functionalism contrasts with computational functionalism [121], which is based on Turing machines and equivalents. There is nothing wrong with Turing machines *per se*; rather they are overused and often not fit for purpose [122]. Although Turing machines can simulate physical, chemical and biological phenomena; they are not really a good model of any of them. Differential functionalism is based on the first-order oracle model of computation in [123]: the idea is to posit an oracle that provides gradients, and to count calls to the oracle rather than count operations on a Turing machine. The first-order oracle model is basic to the field of convex optimization because it is better suited to the problems they tackle than Turing machines. It is also closer to the tools actually used to model nature.

A scientific account of phenomenal experience should provide a bridge between properties of experience and properties of physical systems. It has long been hard to imagine how such a bridge would look, but in fact a candidate falls out naturally from staring long enough at Jacobian matrices. To build a case for Jacobians I need to work through simple examples and for this I turn to Gradland where analysis is easy because the physics is designed for, in fact it is, neural nets. In Gradland it is possible to relate recurrence to the experience of duration, distinguish vivid from obscure ideas, describe the overwhelming

confusion of experiencing the world for the first time, understand how learning relates to experience, and so on. How well the bridge built here holds up for more realistic models, especially of brains, is an open question. Differential functionalism is a minimal exemplar to test, play with, learn from, and build on. I see lots of problems, §A.7, you may see more.

Calculus and linear algebra do not a physical theory make, nor a theory of consciousness. However, the worked examples, here and in [14], do at least suggest that calculus and linear algebra could deserve an analogous role, in future theories of consciousness, to their role in physics and in science in general.

References

- [1] Wilfrid Sellars. Philosophy and the scientific image of man. In *Science, Perception and Reality*, pages 35–78. 1962.
- [2] Ernst Mach. *The analysis of sensations and the relation of the physical to the psychical*. 1914.
- [3] David Balduzzi and Giulio Tononi. Integrated information in discrete dynamical systems: Motivation and theoretical framework. *PLOS Computational Biology*, 4(6):e1000091, 2008.
- [4] Masafumi Oizumi, Larissa Albantakis, and Giulio Tononi. From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0. *PLOS Computational Biology*, 10(5):e1003588, 2014.
- [5] Larissa Albantakis, Leonardo Barbosa, Graham Findlay, Matteo Grasso, Andrew M. Haun, William Marshall, William G. P. Mayner, Alireza Zaemzadeh, Melanie Boly, Bjørn E. Juel, Shuntaro Sasai, Keiko Fujii, Isaac David, Jeremiah Hendren, Jonathan P. Lang, and Giulio Tononi. Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLoS Comput Biol*, 19(10):e1011465, 2023.
- [6] Giulio Tononi and Christof Koch. Consciousness: here, there and everywhere? *Phil Trans R Soc B*, 370:20140167, 2015.
- [7] David Balduzzi, Karl Tuyls, Julien Perolat, and Thore Graepel. Re-evaluating evaluation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [8] David Balduzzi, Marta Garnelo, Yoram Bachrach, Wojciech M Czarnecki, Julien Perolat, Max Jaderberg, and Thore Graepel. Open-ended learning in symmetric zero-sum games. In *International Conference on Machine Learning (ICML)*, 2019.
- [9] Meredith Ringel Morris, Dan Altman, Haydn Belfield, Arthur Goemans, Hasan Iqbal, Ryan Burnell, Iason Gabriel, Samuel Albanie, and Allan Dafoe. Characterizing Jaggedness Aids

- Safety & Usability. ...https://www-cs.stanford.edu/~merrie/papers/jaggedness_preprint.pdf, 2026.
- [10] Eric Schwitzgebel. Borderline consciousness, when it's neither determinately true nor determinately false that experience is present. *Philosophical Studies*, 180:3415–3439, 2023.
- [11] Giulio Tononi. An information integration theory of consciousness. *BMC Neuroscience*, 5:42, 2004.
- [12] Giulio Tononi and Melanie Boly. Integrated information theory: A consciousness-first approach to what exists. ...<https://arxiv.org/abs/2510.25998>, 2025.
- [13] Edwin Abbott Abbott. *Flatland: A Romance of Many Dimensions*. Seeley & Co., 1884.
- [14] David Balduzzi. Minimal Models of Affect, Purpose and Norms. 2026.
- [15] Robert Chis-Ciure and Michael Levin. Cognition all the way down 2.0: neuroscience beyond neurons in the diverse intelligence era. *Synthese*, (206–257), 2025.
- [16] Michael Levin and Richard Watson. Machines all the way up and cognition all the way down: Updating the machine metaphor in biology. *Seminars in Cell and Developmental Biology*, 177-178:103668, 2026.
- [17] Michael Levin and David B Resnik. Mind Everywhere: A Framework for Conceptualizing Goal-Directedness in Biology and Other Domains—Part One. *Biological Theory*, 2026.
- [18] Michael Levin and David B Resnik. Mind Everywhere: A Framework for Conceptualizing Goal-Directedness in Biology and Other Domains—Part Two. *Biological Theory*, 2026.
- [19] Philip Goff. Philosophical zombies. *Scientific American*, 34(3):16–18, 2025.
- [20] David J. Chalmers. *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press, 1996.
- [21] Gottfried Wilhelm Leibniz. *Monadology*. 1714.
- [22] Henri Bergson. The Life and Work of Ravaisson. In *The Creative Mind: An Introduction to Metaphysics*. Dover Publications, 2007.
- [23] Michael Faraday. Experimental Researches in Electricity: Series 19-29. *Phil. trans., 1846-52. Other electrical papers from Roy. inst. proc., and Phil. mag.*, 1855.
- [24] Werner Heisenberg. Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen. *Zeitschrift für Physik*, 33:879–893, 1925.
- [25] William James. *The Meaning of Truth*. Prometheus, 1997.
- [26] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is All You Need. In *NeurIPS*, 2017.
- [27] Aakanksha Chowdhery et al. PaLM: Scaling Language Modeling with Pathways. ...<https://arxiv.org/abs/2204.02311>, 2022.
- [28] James J. Gibson. *The ecological approach to visual perception*. Houghton Mifflin, 1979.
- [29] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, pages 8571–8580, 2018.
- [30] Daniel Beaglehole, Peter Súkeník, Marco Mondelli, and Mikhail Belkin. Average gradient outer product as a mechanism for deep neural collapse. In *Advances in Neural Information Processing Systems*, 2024.
- [31] Suraj Srinivas and François Fleuret. Full-gradient representation for neural network visualization. In *Advances in Neural Information Processing Systems*, 2019.
- [32] Bernard J Baars. *A cognitive theory of consciousness*. Cambridge University Press, 1988.
- [33] Stanislas Dehaene and Lionel Naccache. Towards a cognitive neuroscience of consciousness: basic evidence and a workspace framework. *Cognition*, 79:1–37, 2001.
- [34] George A. Mashour, Pieter Roelfsema, Jean-Pierre Changeux, and Stanislas Dehaene. Conscious Processing and the Global Neuronal Workspace Hypothesis. *Neuron*, 105:776–798, 2020.
- [35] R VanRullen and R Kanai. Deep learning and the global workspace theory. *Trends Neurosci*, 44(9):692–704, 2021.
- [36] Wes Gurnee and Nicholas Sofroniew et al. Verbalizable Representations Form a Global Workspace in Language Models. ...<https://transformer-circuits.pub/2026/workspace/index.html>, 2026.
- [37] Maurice Merleau-Ponty. *Phenomenology of perception*. Routledge, 2013.
- [38] Alfred North Whitehead. *Modes of Thought*. Macmillan, 1938.
- [39] B Spinoza. *The Ethics*. Penguin Classics, 1996.
- [40] R J Bell and P Dean. Atomic vibrations in vitreous silica. *Discussions of the Faraday Society*, 50:55–61, 1970.
- [41] Ashok Litwin-Kumar, Kameron Decker Harris, Richard Axel, Haim Sompolinsky, and L. F. Abbott. Optimal degrees of synaptic connectivity. *Neuron*, 93(5):1153–1164, 2017.
- [42] Alfred North Whitehead. *Essays in Science and Philosophy*. Philosophical Library, 1947.

- [43] Gottfried Wilhelm Leibniz and Antoine Arnauld. The Correspondence between Leibniz and Arnauld. In the version by Jonathan Bennett presented at ...www.earlymoderntexts.com, 1687.
- [44] Gustav Kirchhoff. Ueber die auflösung der gleichungen, auf welche man bei der untersuchung der linearen vertheilung galvanischer strömegeführt wird. *Annalen der Physik und Chemie*, 72(12):497–508, 1847.
- [45] C M Fortuin and P W Kasteleyn. On the random-cluster model: I. Introduction and relation to other models. *Physica*, 57(4):536–564, 1972.
- [46] S N Majumdar and D Dhar. Equivalence between the Abelian sandpile model and the $q \rightarrow 0$ limit of the Potts model. *Physica A*, 185(1–4):129–145, 1992.
- [47] C Bogner and S Weinzierl. Feynman graph polynomials. *Int. J. Mod. Phys. A*, 25:2585–2618, 2010.
- [48] Sepp Hochreiter. Untersuchungen zu dynamischen neuronalen netzen. Master’s thesis, Technische Universität München, 1991.
- [49] Yoshua Bengio, Patrice Simard, and Paolo Frasconi. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 1994.
- [50] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 1997.
- [51] Xavier Glorot and Yoshua Bengio. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2010.
- [52] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [53] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Identity mappings in deep residual networks. In *European Conference on Computer Vision (ECCV)*, 2016.
- [54] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. In *CoRR*, 2016.
- [55] Samuel S. Schoenholz, Justin Gilmer, Surya Ganguli, and Jascha Sohl-Dickstein. Deep information propagation. In *International Conference on Learning Representations (ICLR)*, 2017.
- [56] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tieyan Liu. On layer normalization in the transformer architecture. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [57] Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Damai Dai, Huazuo Gao, Jiang Chang, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhengyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, and Wenfeng Liang. mHC: Manifold-constrained hyper-connections. ...<https://arxiv.org/abs/2512.24880>, 2026.
- [58] Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations (ICLR)*, 2014.
- [59] Jeffrey Pennington, Samuel Schoenholz, and Surya Ganguli. Resurrecting the sigmoid in deep learning through dynamical isometry. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [60] Lechao Xiao, Yasaman Bahri, Jascha Sohl-Dickstein, Samuel S. Schoenholz, and Jeffrey Pennington. Dynamical Isometry and a Mean field Theory of CNNs: How to Train 10,000-Layer Vanilla Convolutional Neural Networks. In *ICML*, 2018.
- [61] W Tarnowski et al. Dynamical isometry is achieved in residual networks in a universal way for any activation function. In *AISTATS*, 2019.
- [62] Walter Scott. *Marmion: A Tale of Flodden Field*. 1808.
- [63] William James. How Two Minds Can Know One Thing. *The Journal of Philosophy, Psychology and Scientific Methods*, 2(7):176–181, 1905.
- [64] Paul Skokowski. Sensing Qualia. *Front System Neurosci*, 16:795405, 2022.
- [65] R L Gregory and J G Wallace. Recovery from Early Blindness: A Case Study. In Paul Tibbetts, editor, *Perception*. Quadrangle/New York Times Book Co., 1969.
- [66] Gerald M Edelman and Giulio Tononi. Reentry and the Dynamic Core: Neural Correlates of Conscious Experience. In Thomas Metzinger, editor, *Neural Correlates of Consciousness*. MIT Press, 2000.
- [67] Geraint Rees, Gabriel Kreiman, and Christof Koch. Neural correlates of consciousness in humans. *Nature Reviews Neuroscience*, 3:261–270, 2002.
- [68] Andrea I. Luppi, Pedro A. M. Mediano, Fernando E. Rosas, John T. O’Brien Negin Holland, Tim D. Fryer, James B. Rowe, David K. Menon, Daniel Bor, and Emmanuel A. Stamatakis. A synergistic core for human brain evolution and cognition. *Nature Neuroscience*, 25:771–782, 2022.
- [69] Andrea I. Luppi, Fernando E. Rosas, Pedro A.M. Mediano, David K. Menon, and Emmanuel A. Stamatakis. Information decomposition and the informational architecture of the brain. *Trends Cogn Sci*, 28(4):352–368, 2024.
- [70] Immanuel Kant. *Critique of Pure Reason*. 1781.
- [71] Adrien Doerig, Aaron Schurger, Kathryn Hess, and Michael H. Herzog. The unfolding argument: Why

- IIT and other causal structure theories cannot explain consciousness. *Conscious Cogn*, 72:49–59, 2019.
- [72] Christof Koch, Moshe Rapp, and Idan Segev. A Brief History of Time (Constants). *Cereb Cortex*, 6(2):93–101, 1996.
- [73] J C Magee. Dendritic integration. *Nature Reviews Neuroscience*, 1:181–190, 2000.
- [74] Nelson Spruston. Pyramidal neurons: dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 3:206–21, 2008.
- [75] René Descartes. *Principles of Philosophy*. 1644.
- [76] Vinod Nair and Geoffrey E. Hinton. Rectified linear units improve restricted boltzmann machines. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 2010.
- [77] Xavier Glorot, Antoine Bordes, and Yoshua Bengio. Deep sparse rectifier neural networks. In *Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011.
- [78] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs). ...<https://arxiv.org/abs/1606.08415>, 2016.
- [79] Noam Shazeer. GLU variants improve transformer. ...<https://arxiv.org/abs/2002.05202>, 2020.
- [80] P C Bush and T J Sejnowski. *The cortical neuron*. Oxford University Press, 1995.
- [81] R Douglas et al. Recurrent excitation in neocortical circuits. *Science*, 269(5226):981–985, 2003.
- [82] Lukas Fessler, Mozes Jacobs, Thomas Fel, Andy Keller, and Sham Kakade. A Unifying View of Attention Sinks: Two Algorithms, Two Solutions. ...<https://arxiv.org/abs/2606.08105>, 2026.
- [83] Kenneth D Harris and Gordon M G Shepherd. The neocortical circuit: themes and variations. *Nature Neuroscience*, 18(2):170–181, 2015.
- [84] Martin Heidegger. *Being and Time*. Harper & Row, 1927.
- [85] Adam H Marblestone, Greg Wayne, and Konrad P Kording. Toward an Integration of Deep Learning and Neuroscience. *Front Comp Neurosci*, 10:94, 2016.
- [86] Y Lecun, L Bottou, Y Bengio, and P Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 1998.
- [87] Kunihiro Fukushima. Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by a Shift in Position. *Biol Cybernetics*, 36:193–202, 1980.
- [88] D H Hubel and T N Wiesel. Receptive fields and functional architecture of monkey striate cortex. *J Physiol*, 195:215–243, 1968.
- [89] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet Classification with Deep Convolutional Neural Networks. In *NeurIPS*, 2012.
- [90] William James. *The Principles of Psychology*. Henry Holt and Company, 1890.
- [91] William G Chase and Herbert A Simon. Perception in chess. *Cognitive Psychology*, 4:55–81, 1973.
- [92] David Balduzzi, Marcus Frean, Lennox Leary, JP Lewis, Kurt Wan-Duo Ma, and Brian McWilliams. The shattered gradients problem: If resnets are the answer then what is the question? In *ICML*, 2017.
- [93] J W Goodman. *Speckle Phenomena in Optics: Theory and Applications*. Roberts & Company Publishers, 2006.
- [94] Donald O Hebb. *Organization of Behavior*. Wiley and Sons, 1949.
- [95] Dennis Goldstein. *Polarized Light*. CRC Press, 2010.
- [96] Nelson Elhage et al. Toy models of superposition, 2022. Transformer Circuits Thread.
- [97] G E Alexander, M R DeLong, and P L Strick. Parallel organization of functionally segregated circuits linking basal ganglia and cortex. *Annu Rev Neurosci*, 9:357–381, 1986.
- [98] F A Middleton and P L Strick. Basal ganglia and cerebellar loops: motor and cognitive circuits. *Brain Res Brain Res Rev*, 31(2-3):236–250, 2000.
- [99] Ann M Graybiel. Habits, rituals, and the evaluative brain. *Annu Rev Neurosci*, 31:359–387, 2008.
- [100] Court Hull and Wade G Regehr. The Cerebellar Cortex. *Annu Rev Neurosci*, 45:151–175, 2022.
- [101] Simona Ginsburg and Eva Jablonka. *The Evolution of the Sensitive Soul: Learning and the Origins of Consciousness*. MIT Press, 2019.
- [102] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.
- [103] Eugene M Izhikevich. Solving the distal reward problem through linkage of STDP and dopamine signaling. *Cereb Cortex*, 17(10):2443–2452, 2007.
- [104] R V Florian. Reinforcement Learning Through Modulation of Spike-Timing-Dependent Synaptic Plasticity. *Neural Computation*, 19:1468–1502, 2007.
- [105] Wulfram Gerstner, Marco Lehmann, Vasiliki Liakoni, Dane Corneil, and Johanni Brea. Eligibility Traces and Plasticity on Behavioral Time Scales: Experimental Support of NeoHebbian Three-Factor Learning Rules. *Front Neural Circuits*, 12, 2018.
- [106] William James. *A Pluralistic Universe*. 1909.
- [107] Benjamin Recht. *The Irrational Decision: How we gave computers the power to choose for us*. Princeton University Press, 2026.
- [108] Alan M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950.

- [109] W R Ashby. *An Introduction to Cybernetics*. Chapman and Hall, 1956.
- [110] Peter Godfrey-Smith. *Complexity and the Function of Mind in Nature*. Cambridge University Press, 1996.
- [111] Gabriel de Tarde. *Monadology and Sociology*. re.Press, 2011.
- [112] David Balduzzi. Deep online convex optimization with gated games. ...<https://arxiv.org/abs/1604.01952>, 2016.
- [113] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115, 2021.
- [114] Janus. Simulators. ...<https://www.lesswrong.com/posts/vJFdjigzmcXMhNTsx/>, 2022.
- [115] Murray Shanahan, Kyle McDonnell, and Laria Reynolds. Role play with large language models. *Nature*, 623:493–498, 2023.
- [116] William James. Does ‘Consciousness’ Exist? *Journal of Philosophy, Psychology, and Scientific Methods*, 1(18):477–491, 1904.
- [117] William James. A world of pure experience. *The Journal of Philosophy, Psychology and Scientific Methods*, 1(20):533–543, 1904.
- [118] Alfred North Whitehead. *Science and the Modern World*. Cambridge University Press, 1926.
- [119] Bertrand Russell. *The Analysis of Matter*. Routledge, 2023.
- [120] Erik C Banks. *The Realistic Empiricism of Mach, James, and Russell: Neutral Monism Reconciled*. Cambridge University Press, 2014.
- [121] Hilary Putnam. Minds and machines. In *Mind, Language, and Reality: Philosophical Papers*. Cambridge University Press, 1975.
- [122] Tim van Gelder. What might cognition be, if not computation? *Journal of Philosophy*, 92(7):345–381, 1995.
- [123] Arkadi S. Nemirovski and D B. Yudin. *Problem Complexity and Method Efficiency in Optimization*. Wiley-Interscience, 1983.
- [124] Wikipedia contributors. Singular value decomposition. ...https://en.wikipedia.org/wiki/Singular_value_decomposition, 2026.
- [125] G W Stewart. On the Early History of the Singular Value Decomposition. *SIAM Review*, 35(4):551–566, 1993.
- [126] Gene Golub and Charles F Van Loan. *Matrix Computations*. Johns Hopkins University Press, 2013.
- [127] Gilbert Strang. *Linear Algebra and Learning from Data*. Wellesley-Cambridge Press, 2019.
- [128] F Wegner. Inverse participation ratio in $2+\epsilon$ dimensions. *Zeitschrift für Physik B Condensed Matter*, 36(209–214), 1980.
- [129] Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *European Signal Processing Conference (EUSIPCO)*, 2007.
- [130] Frank Harary, editor. *Graph Theory and Theoretical Physics*. Academic Press, 1967.
- [131] Béla Bollobás. *Modern Graph Theory*. Springer, 1998.
- [132] Richard P. Stanley. *Enumerative Combinatorics, Vol. 2*. Cambridge University Press, 1999.
- [133] Daniel A Spielman and Shang-Hua Teng. Nearly linear time algorithms for preconditioning and solving symmetric, diagonally dominant linear systems. *SIAM J Matrix Anal Appl*, 35(3):835–885, 2014.
- [134] Tom B Brown et al. Language Models are Few-Shot Learners. ...<https://arxiv.org/abs/2005.14165>.
- [135] Catherine Olsson et al. In-context Learning and Induction Heads. ...<https://arxiv.org/abs/2209.11895>.
- [136] Gautam Reddy. The mechanistic basis of data dependence and abrupt learning in an in-context classification task. In *ICLR*, 2024.
- [137] Anil K Seth and Tim Bayne. Theories of consciousness. *Nature Reviews Neuroscience*, 23:439–452, 2022.
- [138] Patrick Butlin and Robert Long et al. Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. ...<https://arxiv.org/abs/2308.08708>, 2023.
- [139] Robert Lawrence Kuhn. A landscape of consciousness: Toward a taxonomy of explanations and implications. *Progress in Biophysics and Molecular Biology*, 190:28–169, 2024.
- [140] Liad Mudrik, Melanie Boly, Stanislas Dehaene, Stephen M. Fleming, Victor Lamme, Anil Seth, and Lucia Melloni. Unpacking the complexities of consciousness: Theories and reflections. *Neuroscience and Biobehavioral Reviews*, 170:106053, 2025.
- [141] Victor A F Lamme. Towards a true neural stance on consciousness. *Trends Cogn Sci*, 10(11):494–501, 2006.
- [142] D Rosenthal. *Consciousness and Mind*. Oxford University Press, 2006.
- [143] Hakwan Lau. A higher-order Bayesian decision theory of consciousness. In R. Banerjee and B. K. Chakrabarti, editors, *Models of Brain and Mind. Physical, Computational and Psychological Approaches*. Elsevier, 2008.
- [144] Joseph E LeDoux and Richard Brown. A higher-order theory of emotional consciousness. *PNAS*, 114(10):E2016–E2025, 2017.

- [145] R Brown, H Lau, and J E LeDoux. Understanding the Higher-Order Approach to Consciousness. *Trends Cogn Sci*, 23(9):754–768, 2019.
- [146] Axel Cleeremans. The radical plasticity thesis: how the brain learns to be conscious. *Front Psychol*, 2:86, 2011.
- [147] Rafael Harth. Why it’s so hard to talk about Consciousness. ...<https://www.lesswrong.com/posts/NyiFLzSrKfkDW4S7o/why-it-s-so-hard-to-talk-about-consciousness>, 2023.
- [148] David Balduzzi and Giulio Tononi. Qualia: the geometry of integrated information. *PLoS Comput Biol*, 5(8):e1000462, Aug 2009.
- [149] David Spivak. *Category Theory for the Sciences*. MIT Press, 2014.
- [150] C Rovelli. Relational quantum mechanics. *International Journal of Theoretical Physics*, 35:1637–1678, 1996.
- [151] Carlo Rovelli. The relational interpretation. In Olival Freire Jr, Olivier Darrigol, Thiago Hartz, Christian Joas, Alexei Kojevnikov, and Osvaldo Pessoa Jr, editors, *The Oxford Handbook of the History of Quantum Interpretations*. Oxford University Press, 2022.
- [152] Carlo Rovelli. *Helgoland: The Strange and Beautiful Story of Quantum Physics*. Penguin, 2022.
- [153] Emily Adlam and Carlo Rovelli. Information is Physical: Cross-Perspective Links in Relational Quantum Mechanics. *Philosophy of Physics*, 1(1):1–19, 2023.
- [154] David Balduzzi. On the information-theoretic structure of distributed measurements. *EPTCS*, 88:28–42, 2012.
- [155] D Balduzzi. Detecting emergent processes in cellular automata with excess information. In *Advances in Artificial Life: ECAL*. MIT Press, 2011.
- [156] David Balduzzi. Information, learning and falsification. In *Philosophy and Machine Learning workshop, Neural Information Processing Systems (NIPS)*, 2011.
- [157] David Balduzzi. Falsification and Future Performance. In David Dowe, editor, *Algorithmic Probability and Friends: Bayesian Prediction and Artificial Intelligence*, volume 7070 of *LNAI*, pages 65–78. Springer, 2013.
- [158] David Balduzzi. Falsifiable implies Learnable. ...<https://arxiv.org/abs/1408.6618>.
- [159] Adam B Barrett. An integration of integrated information theory with fundamental physics. *Front Psychol*, 5:63, 2014.
- [160] Adam B Barrett. A comment on Tononi & Koch (2015) ‘Consciousness: here, there and everywhere?’. *Phil. Trans. R. Soc. B*, 371(1687):20140198, 2016.
- [161] Adam B. Barrett, Borjan Milinkovic, Pedro A. M. Mediano, Fernando E. Rosas, Daniel Bor, Lionel Barnett, and Anil K. Seth. Integrated information theory: the good, the bad and the misunderstood. ...<https://arxiv.org/abs/2604.11482>, 2026.
- [162] David Balduzzi, Brian McWilliams, and Tony Butler-Yeoman. Neural Taylor Approximations: Convergence and Exploration in Rectifier Networks. In *ICML*, 2017.
- [163] William P Thurston. On proof and progress in mathematics. *Bulletin of the AMS*, 30(2):161–177, 1994.
- [164] P W Bridgman. *Dimensional Analysis*. Yale University Press, 1922.
- [165] George W. Hart. *Multidimensional Analysis: Algebras and Systems for Science and Engineering*. Springer, 1995.
- [166] J C Gibbings. *Dimensional Analysis*. Springer, 2011.
- [167] David Balduzzi and Muhammad Ghifary. Strongly-Typed Recurrent Neural Networks. In *ICML*, 2016.
- [168] David Balduzzi. Strongly-typed agents are guaranteed to interact safely. In *ICML*, 2017.
- [169] Christopher J. Whyte, Andrew W. Corcoran, Jonathan Robinson, Ryan Smith, Rosalyn J. Moran, Thomas Parr, Karl J. Friston, Anil K. Seth, and Jakob Hohwy. On the minimal theory of consciousness implicit in active inference. *Physics of Life Reviews*, 56:4–28, 2026.
- [170] Niels Bohr. *The Philosophical Writings of Niels Bohr: Vol 1, Atomic Theory and the Description of Nature*. Ox Bow Press, 1963.
- [171] Adenauer Casali et al. A Theoretically Based Index of Consciousness Independent of Sensory Processing and Behavior. *Science Translational Medicine*, 5(198), 2013.

A Appendices

A.1 Singular value decomposition

Given $n \times d$ matrix $\mathbf{A}$, the **singular value decomposition (SVD)** is a factorization [124–127]

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \quad (7)$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal, $n \times n$ and $d \times d$ respectively, and $\mathbf{\Sigma}$ is an $n \times d$ matrix with diagonal entries $\sigma_i \geq 0$ for all i , and all off-diagonal entries equal to zero.

The diagonal entries σ_i of $\mathbf{\Sigma}$ are the *singular values* of $\mathbf{A}$. Singular values are ordered so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(n,d)}$. The rank $\text{rk}(\mathbf{A})$, typically denoted r , is the number of *nonzero* singular values.

Geometrically, the SVD says every matrix factorizes into a rotation followed by rescaling and another rotation. It follows that

$$\mathbf{A}\mathbf{v}_i = \sigma_i \mathbf{u}_i \quad \text{and} \quad \mathbf{A}^\top \mathbf{u}_i = \sigma_i \mathbf{v}_i$$

where $\mathbf{u}_i$ and $\mathbf{v}_i$ are orthonormal vectors in $\mathbb{R}^n$ and $\mathbb{R}^d$ respectively, given by the columns of $\mathbf{U}$ and $\mathbf{V}$.

A.2 Effective rank

Let $\mathbf{A}$ be a matrix with rank r , and nonzero singular values $\sigma_1, \dots, \sigma_r$. Then **effective rank** or the **participation ratio**, repeating definition 1, is

$$\rho(\mathbf{A}) := \begin{cases} \frac{(\sum_{i=1}^r \sigma_i^2)^2}{\sum_{i=1}^r \sigma_i^4} & \text{when } \text{rk}(\mathbf{A}) > 0 \\ 0 & \text{else} \end{cases}$$

The inverse participation ratio is also a well-known object. It is the standard diagnostic for Anderson localization in condensed matter physics [128]: it distinguishes an eigenstate spread over the whole system from one concentrated on a few sites.

Litwin-Kumar *et al* define the **dimension of a neural representation** as the participation ratio of the population covariance spectrum (they work with eigenvalues of covariance matrices, which are squared singular values of the factor matrix). They show it is maximized at degrees of synaptic connectivity that match those empirically observed in cerebellar granule cells and *Drosophila* Kenyon cells [41].

For the remainder of this subsection, suppose matrix $\mathbf{A}$ has at least one nonzero singular value. Here are some alternate perspectives on effective rank:

1. *Ratio of traces.* Effective rank can be rewritten as

$$\rho(\mathbf{A}) = \frac{\text{tr}(\mathbf{A}\mathbf{A}^\top)^2}{\|\mathbf{A}\mathbf{A}^\top\|_F^2} = \frac{\text{tr}(\mathbf{A}\mathbf{A}^\top)^2}{\text{tr}((\mathbf{A}\mathbf{A}^\top)^2)}$$

$\mathbf{A}\mathbf{A}^\top$ can be replaced with $\mathbf{A}^\top \mathbf{A}$ in the above without changing the result.

2. *Inverse collision probability.* If the squared singular values are normalized to form a probability distribution,

$$p_i = \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2}$$

then effective rank is

$$\rho(\mathbf{A}) = \frac{1}{\sum_{i=1}^r p_i^2},$$

where the denominator $\sum_{i=1}^r p_i^2$ is the *collision probability*, the probability that two independent draws from distribution p yield the same index.

The collision probability varies between maximum of 1, when there is one nonzero singular value, and minimum of $\frac{1}{r}$, when there are r equal singular values. It follows that, when rank is nonzero, effective rank lies in $[1, r]$.

3. *Coefficient of variation of squared singular values.* If $\mu = \frac{1}{r} \sum_{i=1}^r \sigma_i^2$ and $s^2 = \frac{1}{r} \sum_{i=1}^r (\sigma_i^2 - \mu)^2$ are the mean and variance of the squared singular values respectively, then effective rank is

$$\rho(\mathbf{A}) = \frac{r}{1 + (s/\mu)^2} \quad \text{when } \text{rk}(\mathbf{A}) > 0,$$

where $\frac{s}{\mu}$ is the *coefficient of variation* of the squared singular values.

By this formula, effective rank is rank (the numerator) penalized by unevenness of the squared singular values. If the variance of squared singular values is zero, then effective rank is the rank. As variance increases, effective rank decreases.

An alternative softening of the notion of rank, extremely close to effective rank in spirit, is based on entropy. Assume $\text{rk}(\mathbf{A}) > 0$ and define probability distribution

$$q_i = \frac{\sigma_i}{\sum_{j=1}^r \sigma_j}$$

with entropy

$$H(\mathbf{A}) = - \sum_{i=1}^r q_i \log(q_i).$$

Entropic rank is then

$$\text{ent-rk}(\mathbf{A}) = \begin{cases} \exp(H(\mathbf{A})) & \text{if } \text{rk}(\mathbf{A}) > 0 \\ 0 & \text{else} \end{cases}$$

Entropic rank (which is, confusingly, also known as effective rank in the literature [129]): lies in range $[0, r]$; is maximized when all singular values are equal and nonzero; is approximately equal to one when a single singular value dominates; and so on.

A.3 Cohesiveness and Kirchhoff complexity

Given graph G with no edges from a node to itself. For each pair of nodes $i \neq j$ there is a weighted edge $w_{ij} = w_{ji} \geq 0$. If there is no edge, the weight is zero.

Definition 6. Define the **graph Laplacian matrix** [130–132]

$$L_{ij} = \begin{cases} -w_{ij} & i \neq j \\ \sum_{k \neq i} w_{ik} & i = j \end{cases}$$

The graph Laplacian is a symmetric matrix by construction. At least one of its eigenvalues is necessarily zero since $\mathbf{L}(G) \cdot \mathbf{1} = \mathbf{0}$, also by construction.

A tree is a subgraph of G in which every pair of nodes is connected by at most one path. A spanning tree in G is a tree that goes through all nodes of G . Let $\mathcal{T}(G)$ denote the set of all spanning trees in G .

Theorem 1 (Kirchhoff’s matrix tree theorem [130–132]). *Let $\mathbf{L}'(G)$ be a principal submatrix of a graph Laplacian $\mathbf{L}(G)$, i.e. a matrix obtained by deleting the same index row and column (it doesn’t matter which). Then*

$$\det \mathbf{L}'(G) = \sum_{T \in \mathcal{T}(G)} \prod_{e \in T} w_e$$

is the sum of the product of the weights over all spanning trees of G .

Define Kirchhoff complexity by suitably normalizing the expression in Theorem 1.

Definition 7. For a graph G with n nodes define the **Kirchhoff complexity** as

$$\kappa(G) = \left(\frac{\det(\mathbf{L}'(G))}{n^{n-2}} \right)^{1/(n-1)}$$

Kirchhoff complexity is zero if and only if G is disconnected. That is, G decomposes into two subsets of nodes such that weights of all edges connecting the subsets are zero.

Bipartite graphs from arbitrary matrices. Given $n \times d$ matrix $\mathbf{A}$ with $A_{ij} \geq 0$ for all ij , form $(n+d) \times (n+d)$ symmetric Laplacian matrix with block structure

$$\mathbf{L}(\mathbf{A}) = \begin{pmatrix} \mathbf{D} & -\mathbf{A} \\ -\mathbf{A}^\top & \mathbf{E} \end{pmatrix}$$

where $\mathbf{D}$ and $\mathbf{E}$ are the unique diagonal matrices ensuring $\mathbf{1}^\top \mathbf{L} = \mathbf{0}$ and $\mathbf{L} \mathbf{1} = \mathbf{0}$. Concretely, $D_{ii} = \sum_j A_{ij}$ and $E_{jj} = \sum_i A_{ij}$. The matrix $\mathbf{L}(\mathbf{A})$ is the graph Laplacian of a bipartite graph, with two blocks of n and d nodes respectively; there are edges between but not within the two blocks of nodes.

The bipartite case needs its own normalizing constants. The definition of κ above divides by n^{n-2} , the number of spanning trees of the complete graph K_n on n nodes (Cayley’s formula), and takes an $(n-1)^{\text{st}}$ root because a spanning tree of K_n has $n-1$ edges. For the bipartite graph $G(\mathbf{A})$ there are $n+d$ nodes, a spanning tree has $n+d-1$ edges, and the complete bipartite graph $K_{n,d}$ has $n^{d-1}d^{n-1}$ spanning trees.

Diagonal coherency matrices from cohesive matrices. As a simple exercise, compute singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ of some random matrices $\mathbf{A}$ and inspect $\mathbf{B} := \mathbf{D}\mathbf{V}^\top$, which has diagonal coherency matrix $\mathbf{G} = \mathbf{B}\mathbf{B}^\top = \mathbf{D}^2$. Matrices $\mathbf{B}$ constructed from random samples in this way have relatively high cohesion in general, whereas the corresponding coherency matrices $\mathbf{G}$ have zero cohesion.

The Fiedler eigenvalue. To check whether a graph is connected, it suffices to compute the second smallest eigenvalue of the graph Laplacian, known as the Fiedler eigenvalue. Note the smallest eigenvalue of a graph Laplacian is always zero. In some cases, the Fiedler eigenvalue can be approximated much more efficiently than the computational cost of the determinant of the matrix [133].

Large matrices. Cohesion has the nice equivariance property: $\chi(\alpha \cdot \mathbf{A}) = \alpha \cdot \chi(\mathbf{A})$ for $\alpha > 0$. Equivariance does not come for free; it requires computing the $(n+d-1)^{\text{th}}$ root. For large matrices, i.e. large values of n or d , this drags values of $\chi(\tilde{\mathbf{A}})$ close to 1, and cohesion loses its power to detect *approximate* bottlenecks, where interactions between subsystems are weak but not zero.

A more robust measure for detecting approximate bottlenecks at scale is to undo (or not do) the root and take the logarithm:

$$\Delta(\mathbf{A}) := -(n+d-1) \log \chi(\tilde{\mathbf{A}}) \geq 0. \quad (8)$$

A.4 The experience of symbol manipulation

Transformers can one-shot generalize from patterns in their prompts [134], a capability known as in-context learning. One component of in-context learning is *induction heads* [135]: if pattern AB appears in the prompt then an induction head will respond to a second instance of A by producing B (even if AB never appeared in the training data). Induction heads provide flexible primitives for symbol manipulation and reasoning.

It is interesting to work through the experience of symbol manipulation, perhaps with an LLM:

1. Construct an induction head using a two-layer transformer, for example following [136].
2. Compute the gradients when AB is later followed by A and the expected next output is B .
3. Under suitable assumptions, work through the experience of training on more abstract pairs, such as capital-country, instead of symbol-symbol.

A.5 Theories of consciousness

Differential functionalism is not a theory of consciousness. Nevertheless, it attempts to explain aspects of experience and is relevant to work on consciousness. Useful reviews of theories of consciousness are [137–140]. Some prominent theories are:

- T1. *Global workspace theory* [32–35] proposes that neural states are conscious when they enter the so-called global workspace, which is how information processing performed by more specialized modules is integrated across the brain.
- T2. *Recurrent processing theory* [141] proposes that conscious (primarily visual) perception arises with recurrent processing; feedforward processing is insufficient.
- T3. *Higher-order theories* [142–145] argue that consciousness has to do with first-order states being monitored by higher order states.
- T4. *The radical plasticity thesis* [146] emphasizes learning, and is a form of higher-order theory.
- T5. *Integrated information* [3–5], see §A.6.

Global workspace, recurrent processing, and higher-order theories are oriented towards criteria that determine when cortical processing is and is not conscious. In contrast, differential functionalism and integrated information theory tackle experience at a more fundamental level. In Harth’s terminology [147], the first four theories are in camp #1. Differential functionalism and integrated information belong to camp #2.

If a theory of consciousness were eventually constructed building on differential functionalism, it seems plausible that it would be closely related to the theories above. A global workspace, if it exists, is likely to form a large, dense knot of experience, and would therefore play an important role, see [36] for example. Mammalian brains are recurrent to a far greater extent than RNNs or transformers, which likely implies recurrent processing is *necessary* for a significant fraction of cognition in biological brains – even if it is less important in some approaches to cognition. Finally, any future theory would almost certainly be a higher-order theory because learning is a higher-order experience that is inextricably entangled in brain function.

Differential functionalism does not provide a theory of human consciousness, because that requires understanding human brain function. It provides a workable bridge between models of interactions and experience, that can be adapted and improved. I hope it helps sharpen the proposals above, and numerous others.

A.6 Integrated information

Differential functionalism grew out of dissatisfaction with some technical and conceptual features of integrated information.

- I1. Effective rank is analogous to effective information. It captures the idea that the greater the range of differences that make a difference, the richer the experience. It is exponentially easier to compute at scale than effective information.
- I2. Cohesive rank is analogous to integrated information. It captures the idea that a system that decomposes into non-interacting subsystems is not a system at all. It is doubly exponentially easier to compute at scale than integrated information because integrated information requires computing effective information (first exponential) over all partitions (second exponential).
- I3. The qualia space geometry in [148] is replaced with the geometry of Jacobians which is easier to compute and cleaner to work with. Technically, the difference between the two approaches is that the qualia geometry is built on (inverse images of probabilistic) functions and the Jacobian geometry is built on linear transforms; these are very different categories [149]. Jacobians have an extremely rich structure; the paper merely skimmed the surface. One can analyze them layer by layer, relative to parameters, activations, and mixtures of both. Over different timescales, over different sequences of inputs and outputs, as weights adapt during learning, and so on. Jacobians can be converted into various Gram matrices to understand covariances and correlations between parameters and activations. Low rank approximations to Jacobians, often based on SVD, are a powerful, frequently used window into the structure of nets and their dynamics. There is an enormous literature studying Jacobians and related objects in deep learning.
- I4. The conceptual foundation of the qualia geometry is the idea that *all physical interactions are measurements* [150–153], and measurements, as physical interactions, have internal structure [154]. The same could be said for compositional functionalism.
- I5. The concept of *transparency*, that Jacobians are like points of view and that points of view can be composed, is in principle compatible with the formalism of integrated information, but was never properly developed within that framework, although see [155] for awkward steps in this direction.

16. A bridge from effective information to learning theory is provided in [156, 157]: the effective information generated by empirical risk minimization is a simple function of the Vapnik-Chervonenkis entropy, and so provides bounds on generalization. More generally, effective information allows to straightforwardly reformulate results from statistical learning theory, sequential prediction (online learning) and Solomonoff’s universal induction in terms of falsifying hypotheses [158]: the more effective information your learning algorithm generates in each setting, the more hypotheses it falsifies, and the stronger the generalization bounds that hold.
17. The results in [3] on various toy systems can be reproduced using cohesive rank. For example, modular systems have low integrated information and also have low cohesive rank. Modularity manifests as approximately block diagonal Jacobians (perhaps after some permuting), and these have low cohesion. I have not presented examples of modularity yielding low cohesion since it is trivial and would also be somewhat artificial: no-one (?) builds neural nets that are modular in this way. Similarly for systems with saturated activity, which correspond to toy models of seizures.
18. Barrett hypothesized that consciousness arises from the integrated information generated by fundamental fields [159–161]. Reimagining information integration in the language of gradients and matrices is a step in this ambitious direction.

A.7 Open problems

- O1. The specific measures proposed, effective rank and Kirchhoff complexity for cohesion, are fairly arbitrary. The normalization chosen for cohesion, dividing by $\text{mean}_{i,j} |A_{ij}|$ instead of max or root-mean-square, has strong implications for its behavior for large matrices, for example on large sparse versus large Gaussian matrices, and may not be ideal. For highlighting bottlenecks and blind spots in particular, as opposed to measuring interwovenness on average, the Fiedler eigenvalue or Eq. (8) are better starting points. Lots of slightly, or even very, different choices make sense. Hopefully examples will be constructed or discovered that constrain the set of possibilities. Having said that, there are lots of questions one can ask about experience, and there is no reason to expect a small set of tools to answer them all.

It seems important to have both scale-invariant (transparency) and scale-equivariant (luminosity) measures. Scale-invariance is useful when looking at the richness of internal structure, but by definition it is blind to a matrix uniformly ‘fading away’. Scale-equivariance understands when a matrix ‘fades away’, but by definition is easy to juice by rescaling.

- O2. Gradients might not be the whole story, or even part of the story. You could work with first-order Taylor approximations [162]. That is, keep track of the zeroth order (constant) terms as well as the first order (gradient) terms. You could use finite differences or symbolic differentiation. There are lots of ways of thinking about derivatives and the kind of window they provide into the structure of functions [163], and correspondingly lots of options. You could even use information theory.
- O3. Second-order terms appear in the experience of learning. Could they, and higher order terms, be important generally? Yes, especially since biological systems continually modify their internal structure across a variety of spatiotemporal scales, wheels within wheels. The downside is that Hessians and friends are expensive.
- O4. Physical quantities come with units: meters, seconds, joules, grams. This is not a problem for computing Jacobians, everything will just work. However, it is a problem for the singular value decomposition, luminosity, effective rank, and cohesion which do not know that different entries of the Jacobian can have different units. See [164–166] for some discussion of linear algebra with units.

Does experience have units? Hypothesis G is that all physical interactions count. It is logically coherent to suppose only some physical interactions matter, and to pick these out based on their units. However this is done, it probably needs to play well with matrix multiplication (composition of transforms).

Even if all interactions count, it is possible that the units of the components of experience are relevant to what experience is like. That is, that the units of the entries of the Jacobian are relevant to qualia.

- O5. It is interesting to speculate about how change of basis is experienced, and what simultaneous diagonalizability or commutativity implies about experience [167, 168].

- O6. The framework can and should be extended to cover probabilistic systems. For example, partially observable Markov decision processes (POMDPs) are built on linear algebra and probability theory [169].
- O7. Similarly for continuous time.
- O8. What spatiotemporal scale to work at? Are the nodes in the graph at the level of atoms, cells, or something else? What are their inputs? How are they perturbed? The only answers I can imagine here are empirical: the relevant scales and perturbations are those that map best to our lived experience. This is vague, and cashing it out seems hard.
- O9. In Gradland there are artificial neurons with parameters that are updated periodically. Biological neurons are more complicated. It may not be possible to separate out parameters from other structural features, all of which are constantly changing to varying degrees.
- O10. The results on vividness in §4.3 assume that gradients, and gradients alone, are how interactions are experienced. If they are not, if hypothesis G needs extending to zeroth or second order terms say, the picture could change dramatically.
- O11. Boundaries and specifying systems are important of course, §3.3. It should be possible to describe how boundaries between systems form (where I end and you begin, say), when they form, and explain why they do not if and when they do not. In mammalian brains: what the system is, its inputs, its parameters, its outputs. The answers are likely to be empirical and context-dependent.

“One need only remember here the sensation, often cited by psychologists, which every one has experienced when attempting to orient himself in a dark room with a stick. When the stick is held loosely, it appears to the sense of touch to be an object. When, however, it is held firmly, we lose the sensation that it is a foreign body, and the impression of touch becomes immediately localized at the point where the stick is touching the body under investigation.” —Bohr [170]

A.8 Falsifiability

Falsifiable predictions can be extracted from Hypothesis G by combining it with concrete models of brain function. Examples include: how long experiences endure, which aspects of experience are vivid and obscure, salience of losses or errors or objectives, and the relation between distinct ideas and cortical

circuits. The models would need to be (mostly) differentiable. If they are not, then additional work would be required, see O2, O6 and O7 of §A.7 for example.

A possible blocker is if the measures cannot be made determinate. Effective rank and cohesion are only defined once inputs and outputs are specified. If reasonable choices of boundary and timescale give wildly divergent answers for the same system then the framework does not have determinate content.

A crucial test is whether the behavior of cohesive rank tracks states where experience is agreed to reduce. Anaesthesia, certain sleep stages, and seizures are states where experience is uncontroversially diminished or abolished, and where a measure already exists – the perturbational complexity index – against which cohesive rank can be compared [171]. A prediction is that cohesive rank, computed on an estimated Jacobian, will drop under anaesthesia and correlate with existing measures across sleep stages and seizures.